

TruckDrive box-surface evidence atlas

21 object-frames / 14 object identities / 20 distinct captures

Main evidence: H01, F01 and F03 (3 objects, 4 frames). N04 far is supplementary; all weaker controls remain.

H01 scene_35_31 / Vehicle:97 | pages 2, 3, 4
F01 scene_36_27 / Vehicle:3 | pages 5, 6
F03 scene_36_39 / Vehicle:10 | pages 7, 8, 9, 10, 11
N04 scene_28_12 / Vehicle:11 | pages 12, 13, 14
F04 scene_36_39 / Vehicle:1 | pages 15
F02 scene_36_27 / Vehicle:10 | pages 16, 17
H02 scene_42_8 / Vehicle:87 | pages 18
N01 scene_28_9 / Vehicle:6 | pages 19, 20
N02 scene_28_9 / Vehicle:8 | pages 21, 22
N03 scene_28_12 / Vehicle:9 | pages 23, 24
C01 scene_36_39 / RoadObstruction:5 | pages 25, 26, 27
C02 scene_28_9 / RoadObstruction:22 | pages 28
C05 scene_28_9 / RoadObstruction:18 | pages 29, 30
C06 scene_36_39 / RoadObstruction:7 | pages 31

Existing scientific figures only. English captions are translated descriptions; originals and fixed rows remain unchanged.

Face clearance is not center error. Publisher-release identity, timing/compensation and calibration remain unresolved.

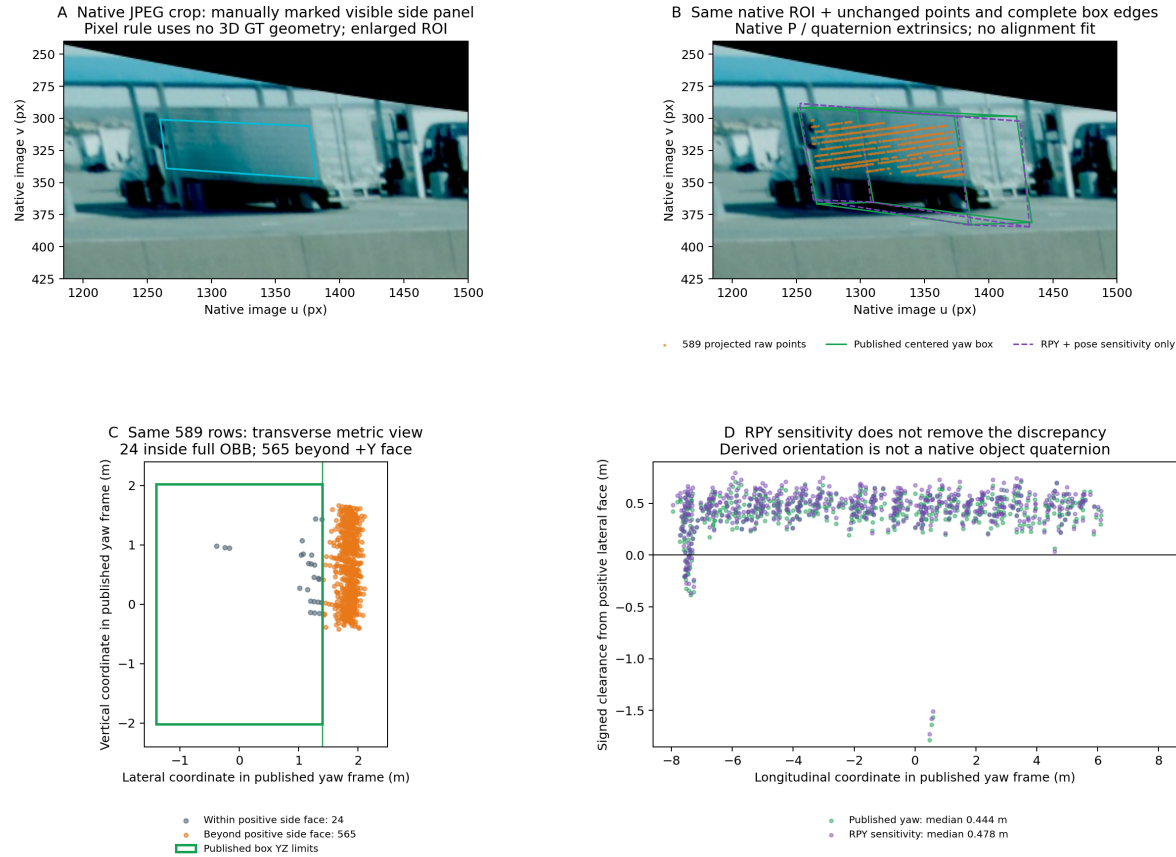
Page repetition does not add cases. See the report and correspondence table for source hashes and precise limitations.

H01 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

scene_35_31 | sync0188 | Vehicle:97 | Tracking_ID 97 | sample 100556 | range 96.99 m

Visible sidewall pixels, released yaw box, RPY sensitivity and existing face measurements

H01 | scene_35_31 / sync0188 / Vehicle:97 / Trailer / range 96.990 m
Single-case geometry audit: point / released-box inconsistency; unique cause remains unresolved



Source: 3848 x 2168 JPEG, float64 Aeva joint scan, native calibration and centered 16.0 x 2.8 x 4.040 m yaw box.
589 pixel-selected rows all have sensor_id=5; physical head mapping and per-point compensation contract are unverified. No detector output or box correction.

Fixed N=589; beyond selected face N=565; +Y side median clearance +0.443503 m. Not center error.

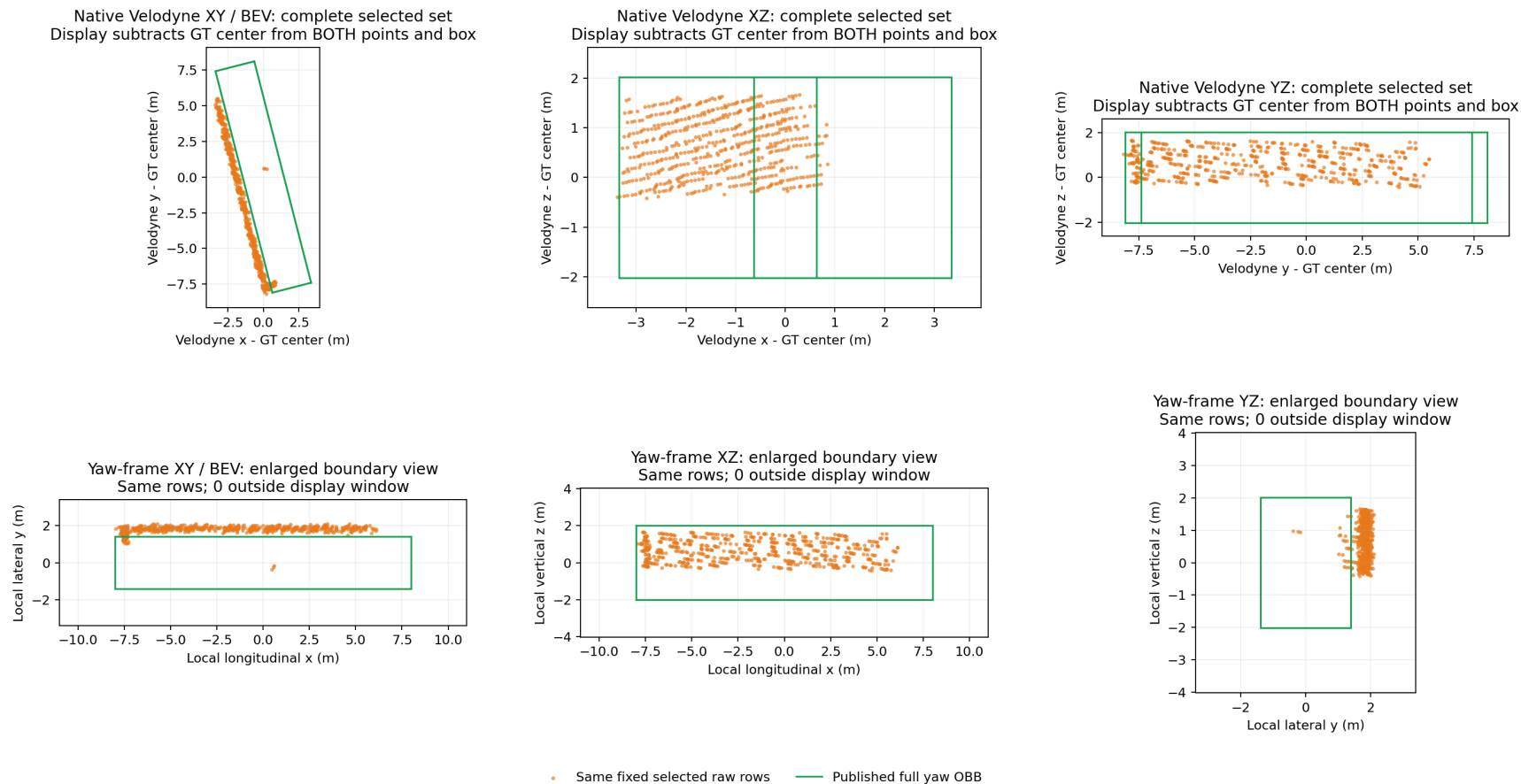
Green is the formal yaw box; purple is sensitivity only. The 589-point pixel rule does not depend on 3D GT geometry; selection was post hoc manual, not blinded.

H01 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

scene_35_31 | sync0188 | Vehicle:97 | Tracking_ID 97 | sample 100556 | range 96.99 m

Camera-independent XY/BEV, XZ, YZ and labeled local enlargement

H01_NATIVE | scene_35_31 / sync0188 / Vehicle:97 | range 96.99 m
Camera-independent orthographic evidence | SAME fixed N=589 | native geometric center / dimensions / yaw



No perspective/camera calibration enters these orthographic views. Upper row includes all fixed rows, including distant background/outliers; lower row only changes the view window. No time attribution or GT correction.

Fixed N=589; beyond selected face N=565; +Y side median clearance +0.443503 m. Not center error.

Joint translation/rotation of boxes and points is for display only. The existing point set is retained; no compensation or fit is applied.

H01 / Vehicle-SemiTruck-Trailer / Inconclusive / projection question / partial consistency

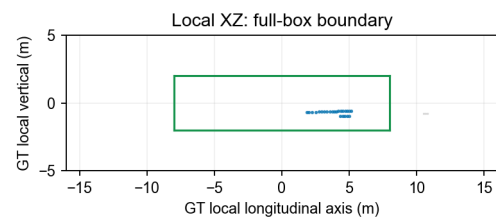
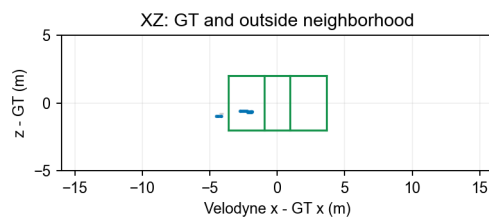
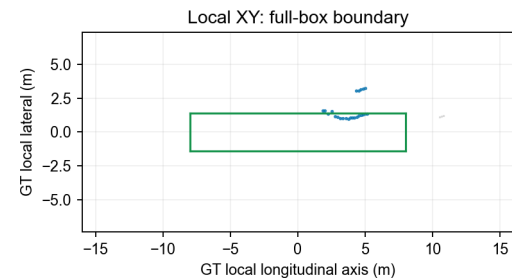
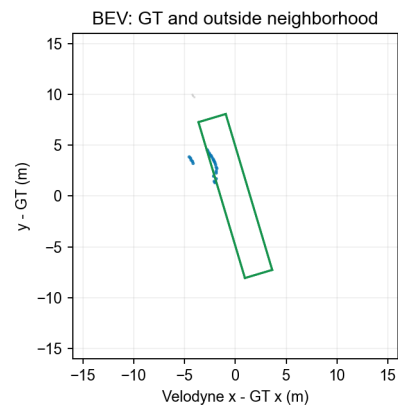
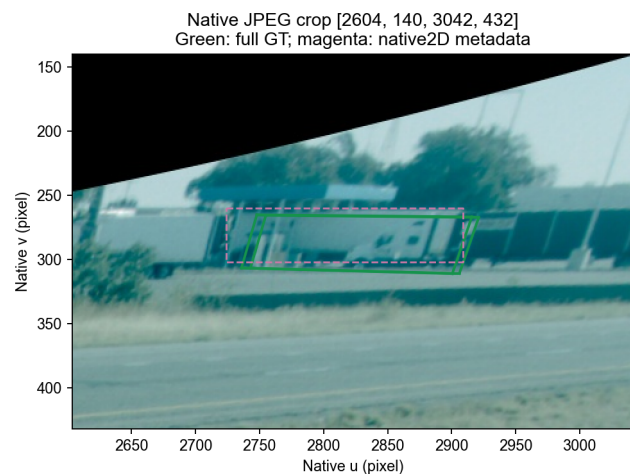
scene_35_31 | sync0129 | Vehicle:97 | Tracking_ID 97 | sample 100497 | range 199.96 m

Native camera, full GT and unassigned historical search returns

H01 | scene_35_31 / sync0129 / Vehicle:97 | Vehicle-SemiTruck-Trailer

Range 199.96 m | GT OBB points 16 | Native-ray local points 24; outside GT 8

FAR | HISTORICAL_IDENTITY_FAR_PAIR | sideward_left_front_wide



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 16. Sparse/mixed/unassigned returns do not establish GT parameter error.

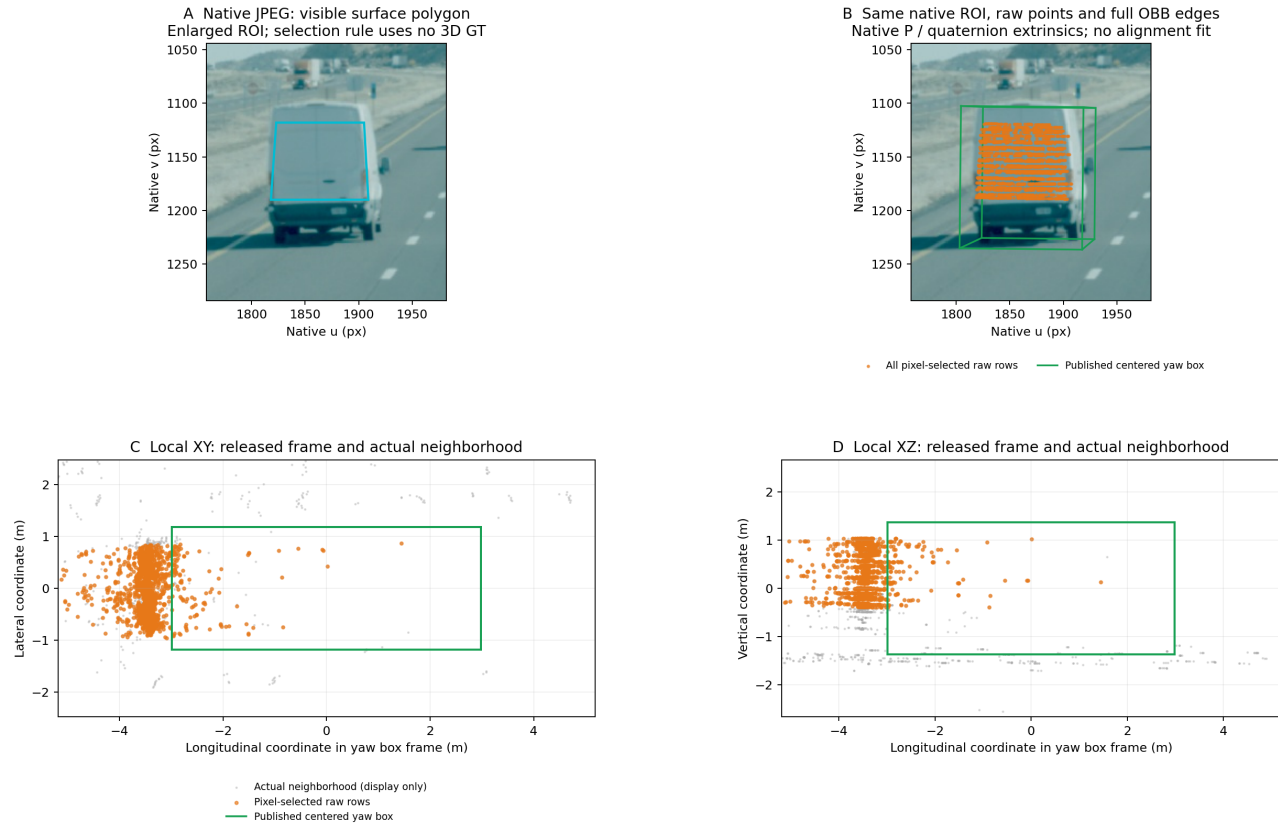
Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

F01 / Vehicle-Passenger / Primary: box-surface discrepancy

scene_36_27 | sync0118 | Vehicle:3 | Tracking_ID 3 | sample 100685 | range 74.88 m

Native ROI, fixed pixel-selected returns, released yaw box and XY/XZ

F01_NEAR_MID | scene_36_27 / sync0118 / Vehicle:3 / Vehicle-Passenger
Range 74.88 m | pixel-selected N=1461, beyond selected face=1351, full OBB inside=109
signed face gap median 0.461 m, P05-P95 -0.124 to 1.061 m



Sensor statistics: id1: N=592, median 0.470 m, id2: N=869, median 0.459 m. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 5 (XY) / 5 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

Fixed N=1461; beyond selected face N=1351; -X rear median clearance +0.461251 m. Not center error.

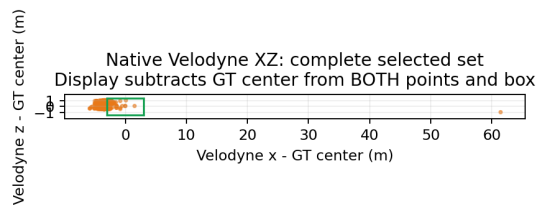
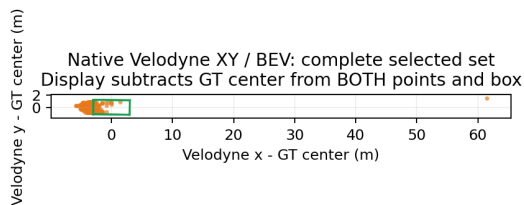
All existing selection rows and tail/outlier returns are retained. Metric face clearance is not object-center error.

F01 / Vehicle-Passenger / Primary: box-surface discrepancy

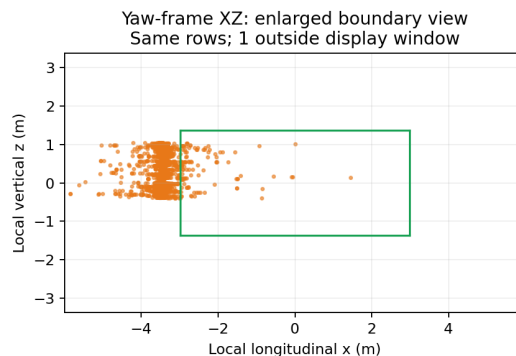
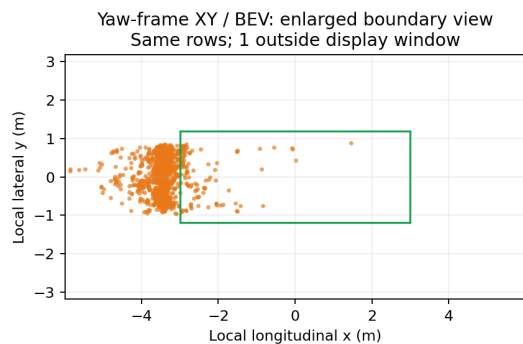
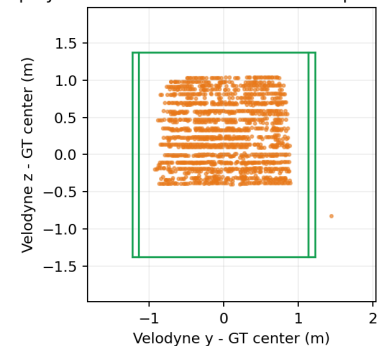
scene_36_27 | sync0118 | Vehicle:3 | Tracking_ID 3 | sample 100685 | range 74.88 m

Camera-independent XY/BEV, XZ, YZ and labeled local enlargement

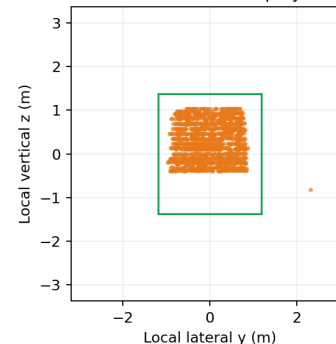
F01_NEAR_MID | scene_36_27 / sync0118 / Vehicle:3 | range 74.88 m
Camera-independent orthographic evidence | SAME fixed N=1461 | native geometric center / dimensions / yaw



Native Velodyne YZ: complete selected set
Display subtracts GT center from BOTH points and box



Yaw-frame YZ: enlarged boundary view
Same rows; 0 outside display window



• Same fixed selected raw rows — Published full yaw OBB

No perspective/camera calibration enters these orthographic views. Upper row includes all fixed rows, including distant background/outliers; lower row only changes the view window. No time attribution or GT correction.

Fixed N=1461; beyond selected face N=1351; -X rear median clearance +0.461251 m. Not center error.

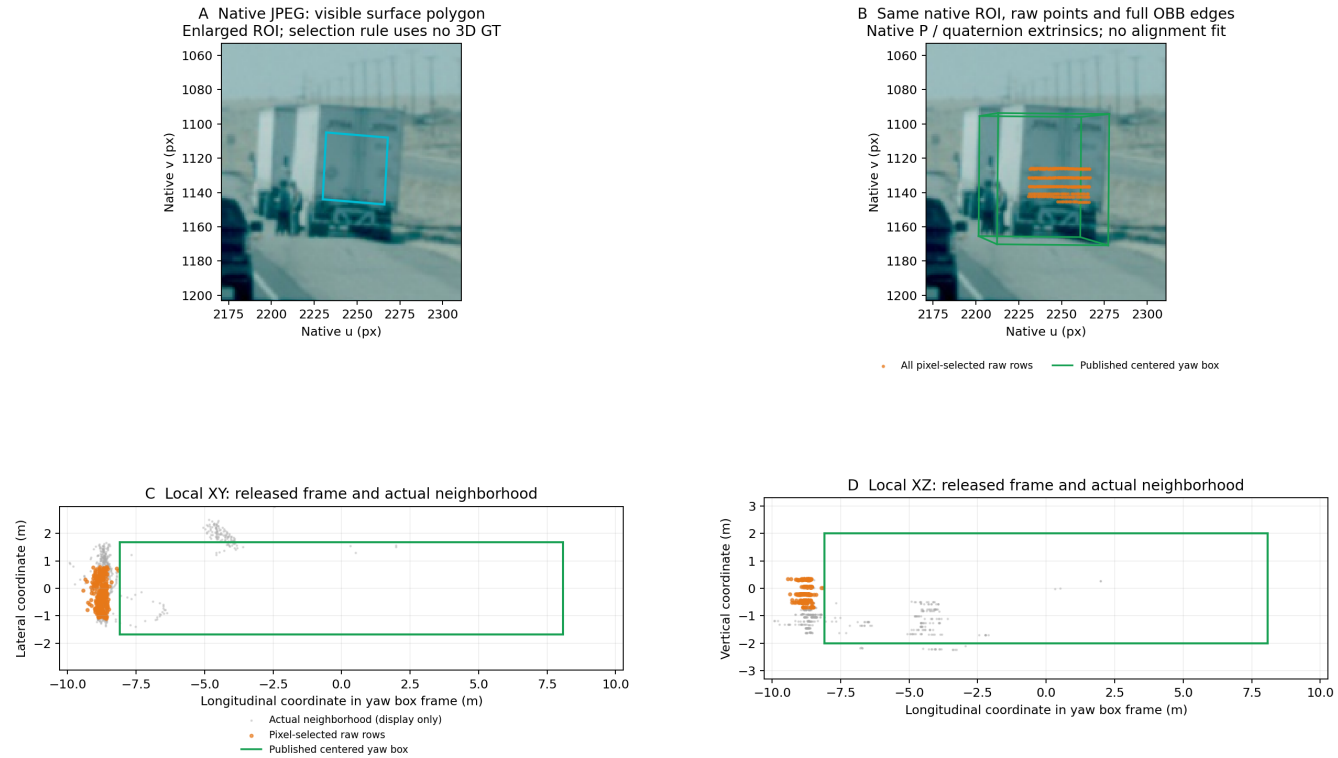
Joint translation/rotation of boxes and points is for display only. The existing point set is retained; no compensation or fit is applied.

F03 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

scene_36_39 | sync0032 | Vehicle:10 | Tracking_ID 10 | sample 100745 | range 186.12 m

Native ROI, fixed pixel-selected returns, released yaw box and XY/XZ

F03_FAR | scene_36_39 / sync0032 / Vehicle:10 / Vehicle-SemiTruck-Trailer
Range 186.12 m | pixel-selected N=282, beyond selected face=282, full OBB inside=0
signed face gap median 0.686 m, P05-P95 0.477 to 0.981 m



Sensor statistics: id1: N=149, median 0.653 m, id2: N=133, median 0.705 m. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 0 (XY) / 0 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

Fixed N=282; beyond selected face N=282; -X rear median clearance +0.685512 m. Not center error.

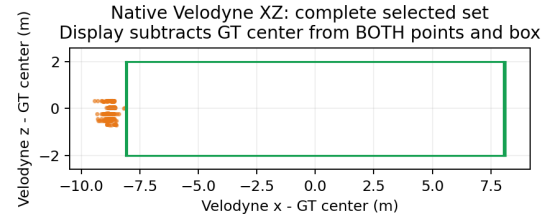
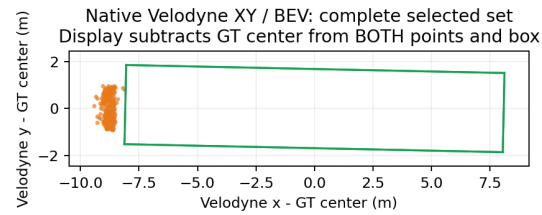
All existing selection rows and tail/outlier returns are retained. Metric face clearance is not object-center error.

F03 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

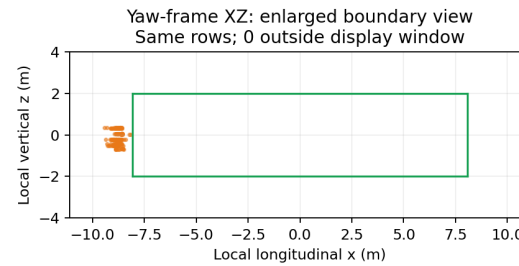
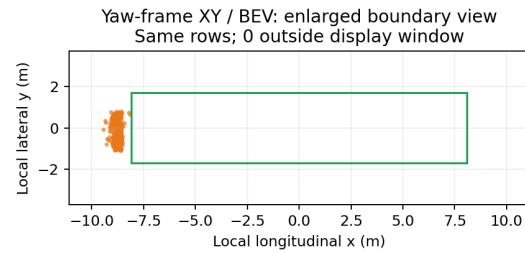
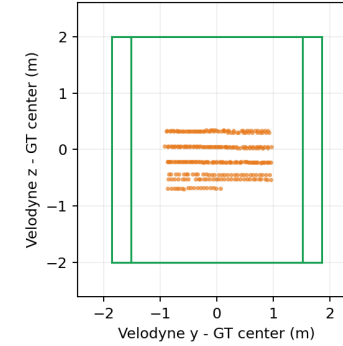
scene_36_39 | sync0032 | Vehicle:10 | Tracking_ID 10 | sample 100745 | range 186.12 m

Camera-independent XY/BEV, XZ, YZ and labeled local enlargement

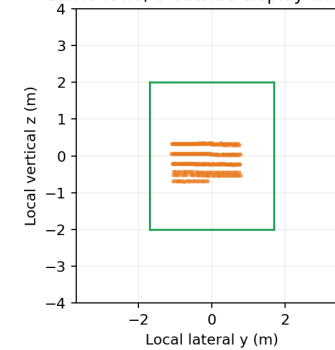
F03_FAR | scene_36_39 / sync0032 / Vehicle:10 | range 186.12 m
Camera-independent orthographic evidence | SAME fixed N=282 | native geometric center / dimensions / yaw



Native Velodyne YZ: complete selected set
Display subtracts GT center from BOTH points and box



Yaw-frame YZ: enlarged boundary view
Same rows; 0 outside display window



• Same fixed selected raw rows — Published full yaw OBB

No perspective/camera calibration enters these orthographic views. Upper row includes all fixed rows, including distant background/outliers; lower row only changes the view window. No time attribution or GT correction.

Fixed N=282; beyond selected face N=282; -X rear median clearance +0.685512 m. Not center error.

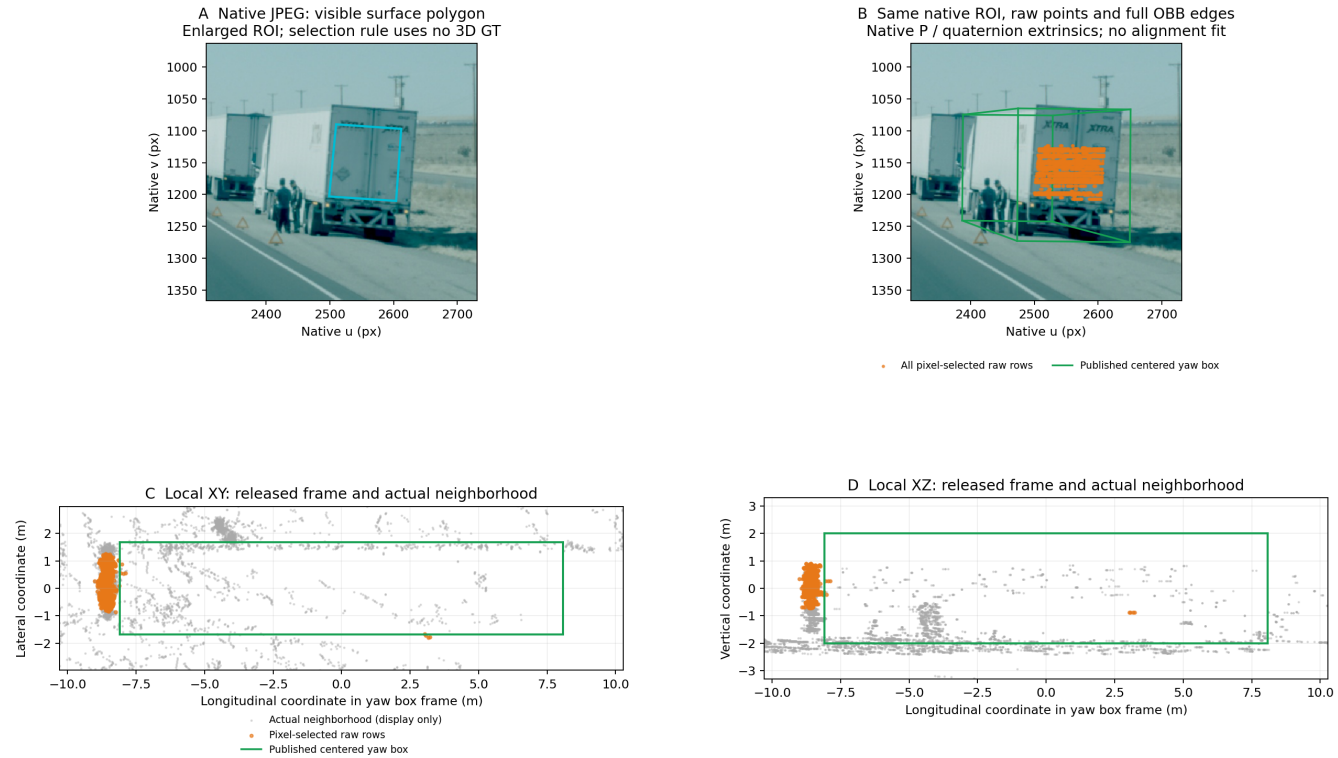
Joint translation/rotation of boxes and points is for display only. The existing point set is retained; no compensation or fit is applied.

F03 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

scene_36_39 | sync0070 | Vehicle:10 | Tracking_ID 10 | sample 100783 | range 76.30 m

Native ROI, fixed pixel-selected returns, released yaw box and XY/XZ

F03_NEAR_MID | scene_36_39 / sync0070 / Vehicle:10 / Vehicle-SemiTruck-Trailer
Range 76.30 m | pixel-selected N=1600, beyond selected face=1592, full OBB inside=5
signed face gap median 0.472 m, P05-P95 0.298 to 0.673 m



Sensor statistics: id1: N=760, median 0.469 m, id2: N=840, median 0.476 m. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 0 (XY) / 0 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

Fixed N=1600; beyond selected face N=1592; -X rear median clearance +0.472415 m. Not center error.

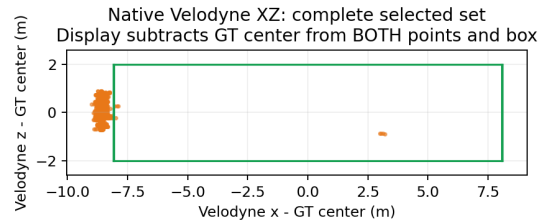
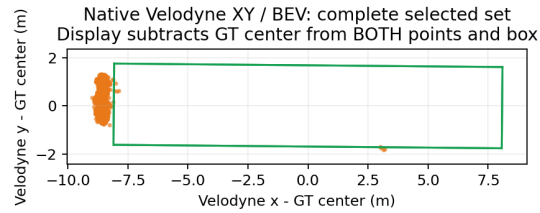
All existing selection rows and tail/outlier returns are retained. Metric face clearance is not object-center error.

F03 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

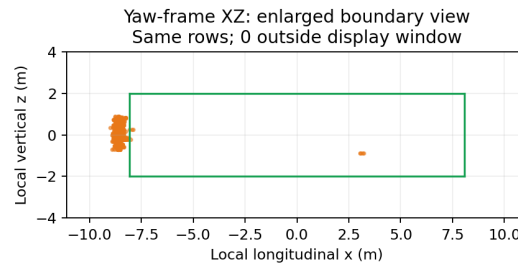
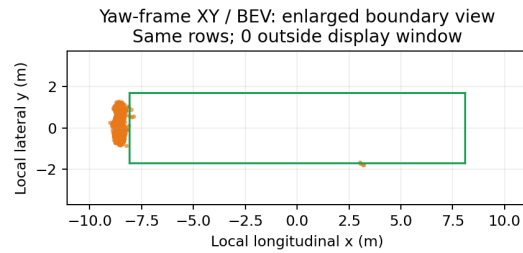
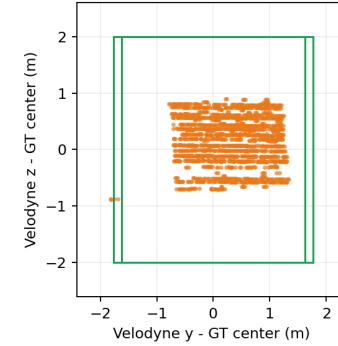
scene_36_39 | sync0070 | Vehicle:10 | Tracking_ID 10 | sample 100783 | range 76.30 m

Camera-independent XY/BEV, XZ, YZ and labeled local enlargement

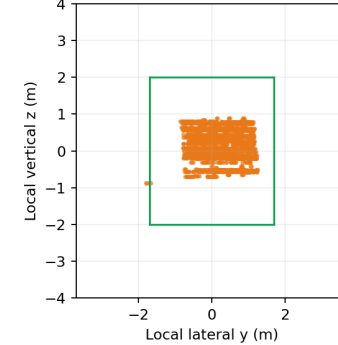
F03_NEAR_MID | scene_36_39 / sync0070 / Vehicle:10 | range 76.30 m
Camera-independent orthographic evidence | SAME fixed N=1600 | native geometric center / dimensions / yaw



Native Velodyne YZ: complete selected set
Display subtracts GT center from BOTH points and box



Yaw-frame YZ: enlarged boundary view
Same rows; 0 outside display window



• Same fixed selected raw rows — Published full yaw OBB

No perspective/camera calibration enters these orthographic views. Upper row includes all fixed rows, including distant background/outliers; lower row only changes the view window. No time attribution or GT correction.

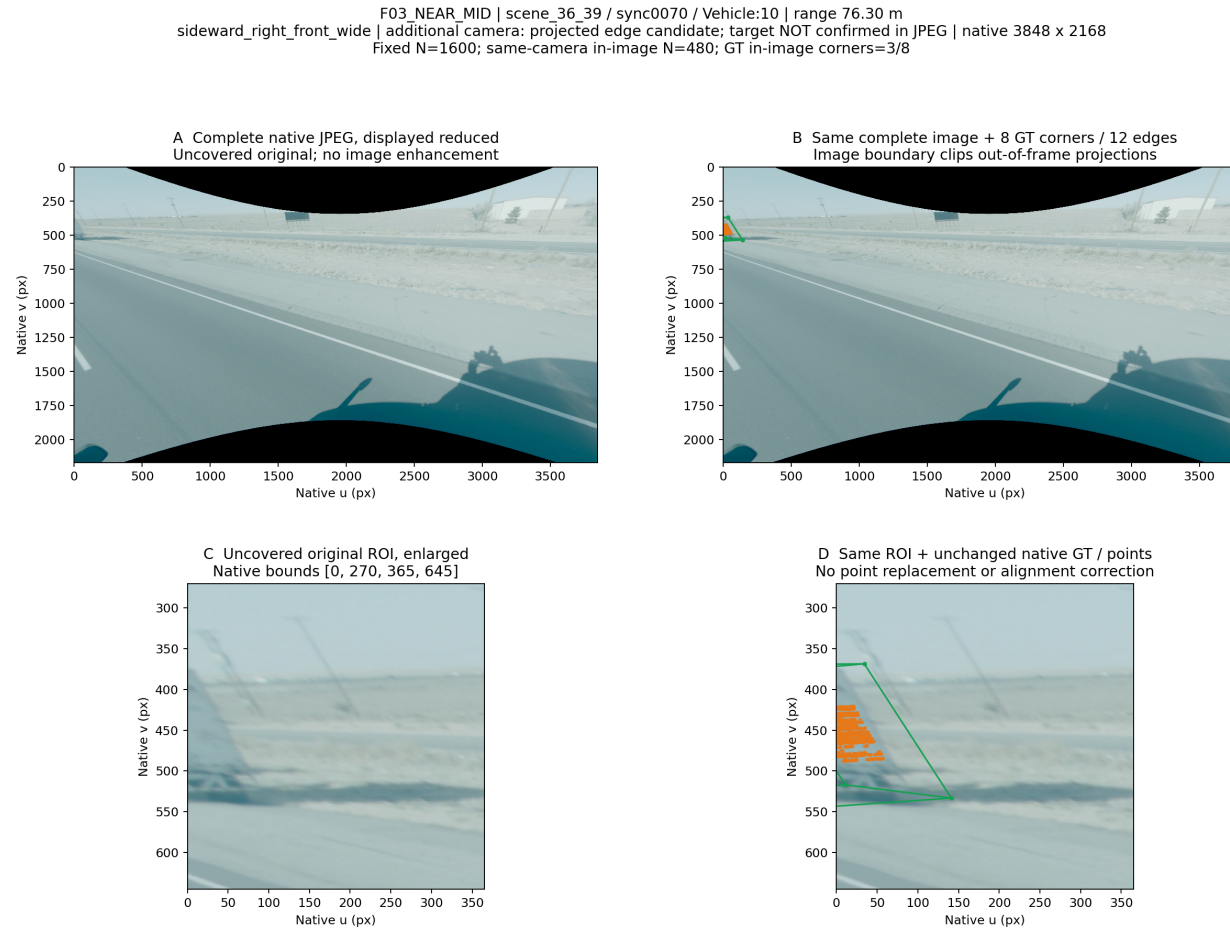
Fixed N=1600; beyond selected face N=1592; -X rear median clearance +0.472415 m. Not center error.

Joint translation/rotation of boxes and points is for display only. The existing point set is retained; no compensation or fit is applied.

F03 / Vehicle-SemiTruck-Trailer / Primary: box-surface discrepancy

scene_36_39 | sync0070 | Vehicle:10 | Tracking_ID 10 | sample 100783 | range 76.30 m

Right-front candidate camera: independent surface view N/A



Green: full published yaw OBB projection. Orange: the original fixed selected rows, including any mixtures/outliers. Clean panels are raw JPEG crops.

Same sync can have different camera timestamp fields. No time-shift claim, compensation, new selection, GT change or Library upload.

— Published native 7D GT • Same fixed raw-point set

Fixed N=1600; beyond selected face N=1592; -X rear median clearance +0.472415 m. Not center error.

480 points and 3 corners enter at the extreme left edge. The clean JPEG does not confirm the same rear door; this is not a confirmed second view.

N04 / Vehicle-SemiTruck-Cab / Supplement: moderate attribution

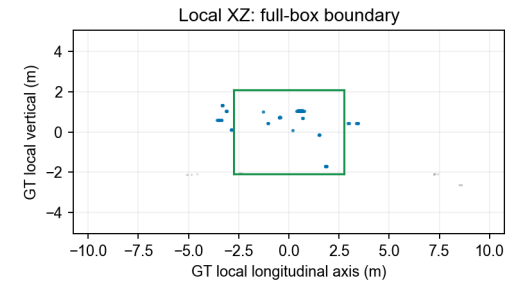
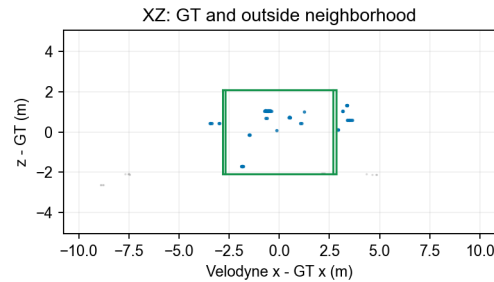
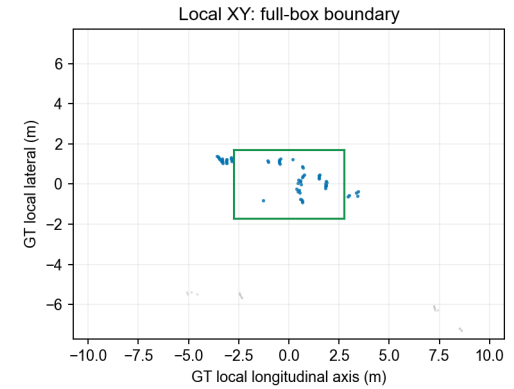
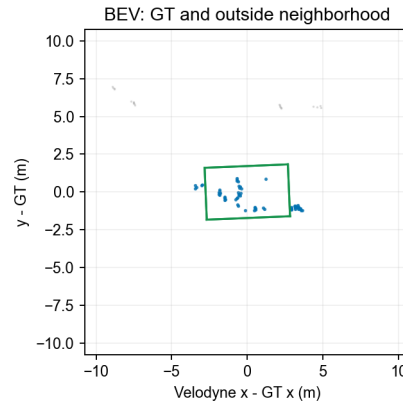
scene_28_12 | sync0062 | Vehicle:11 | Tracking_ID 11 | sample 100000 | range 181.25 m

Native camera, GT and historical search returns

N04 | scene_28_12 / sync0062 / Vehicle:11 | Vehicle-SemiTruck-Cab

Range 181.25 m | GT OBB points 39 | Native-ray local points 64; outside GT 25

FAR | FIXED_CACHED_DEV_METADATA | forward_center_medium



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

GT-dependent corridor: 57 points; 25 beyond end face; max +0.840725 m, overall median -1.258017 m. Moderate attribution.

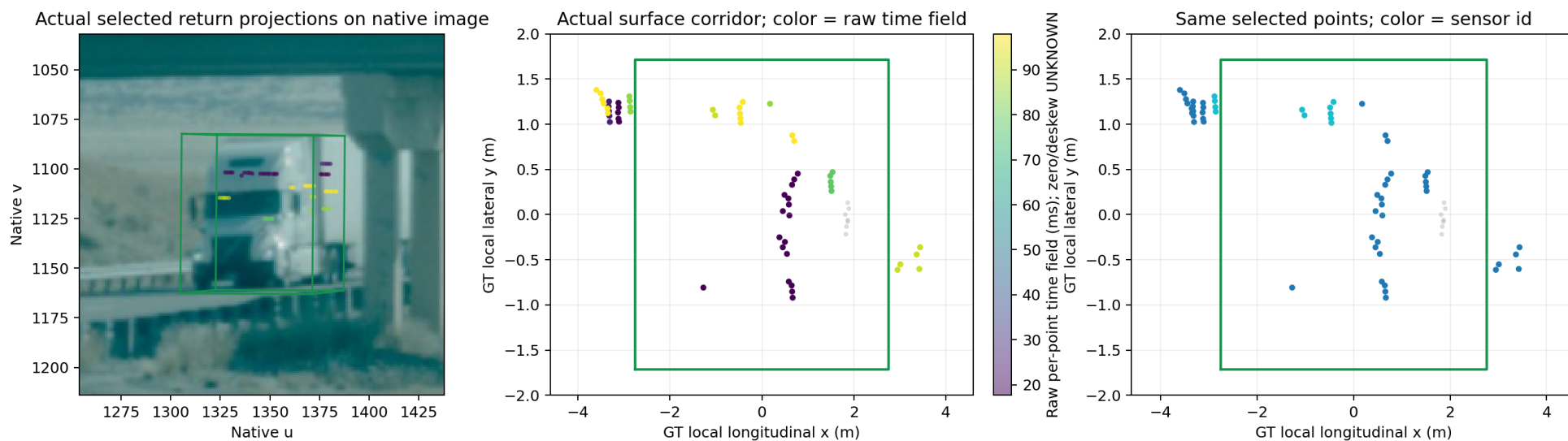
The corridor depends on GT; physical attribution is moderate. A largest tail point is not whole-surface displacement.

N04 / Vehicle-SemiTruck-Cab / Supplement: moderate attribution

scene_28_12 | sync0062 | Vehicle:11 | Tracking_ID 11 | sample 100000 | range 181.25 m

Existing 57-point corridor and end-face details

N04 FAR | real surface-to-GT boundary diagnostic



Selected 57 returns; 25 outside measured GT face. No box fit/translation. Not a center-error estimate.

Shape and actual projection support attribution; not every native2D ray return is assigned. Raw time fields are not certified time relative to GT.

GT-dependent corridor: 57 points; 25 beyond end face; max +0.840725 m, overall median -1.258017 m. Moderate attribution.

25/57 cross the end face; maximum 0.840725 m, overall median -1.258017 m. This is not center error.

N04 / Vehicle-SemiTruck-Cab / Inconclusive / projection question / partial consistency

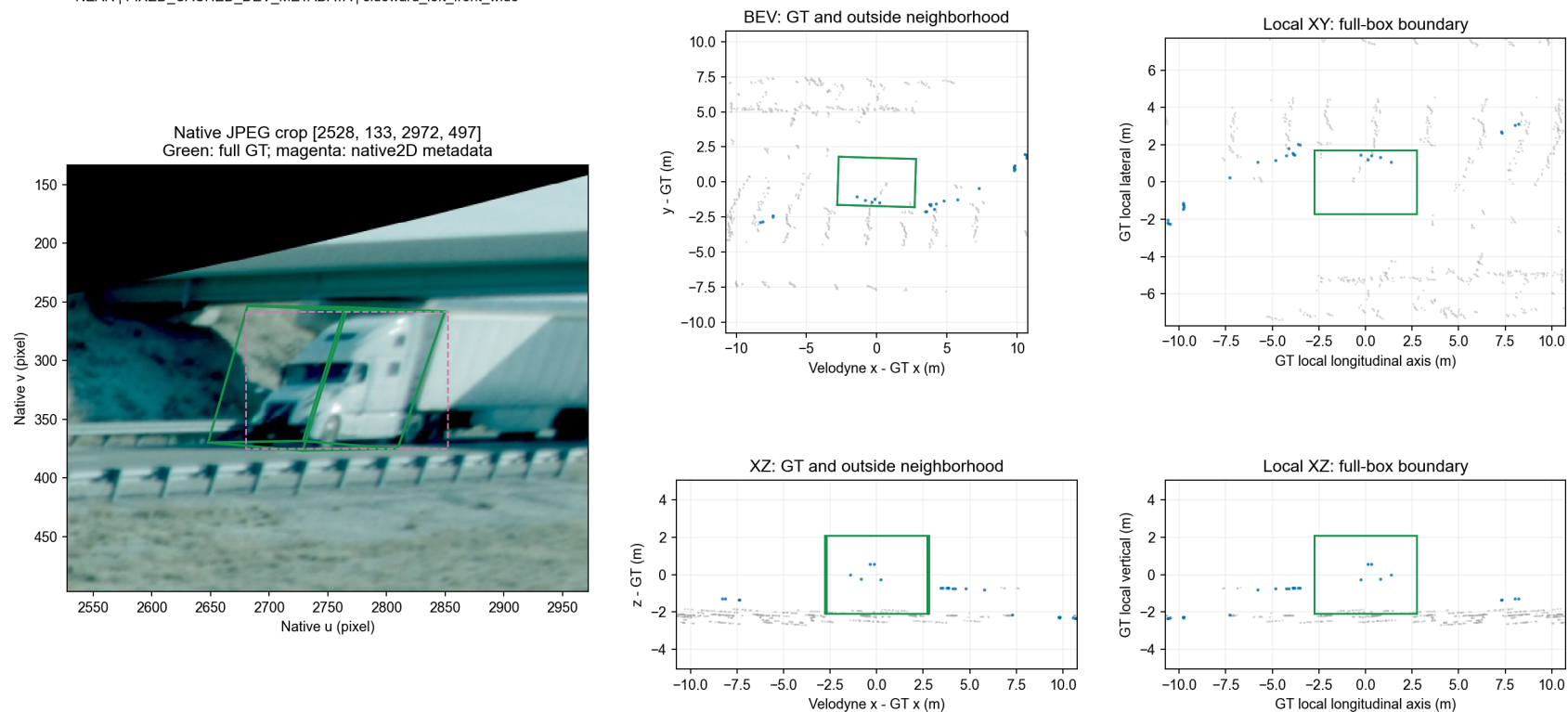
scene_28_12 | sync0100 | Vehicle:11 | Tracking_ID 11 | sample 100019 | range 73.92 m

Native camera, full GT and unassigned historical search returns

N04 | scene_28_12 / sync0100 / Vehicle:11 | Vehicle-SemiTruck-Cab

Range 73.92 m | GT OBB points 5 | Native-ray local points 28; outside GT 23

NEAR | FIXED_CACHED_DEV_METADATA | sideward_left_front_wide



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 5. Sparse/mixed/unassigned returns do not establish GT parameter error.

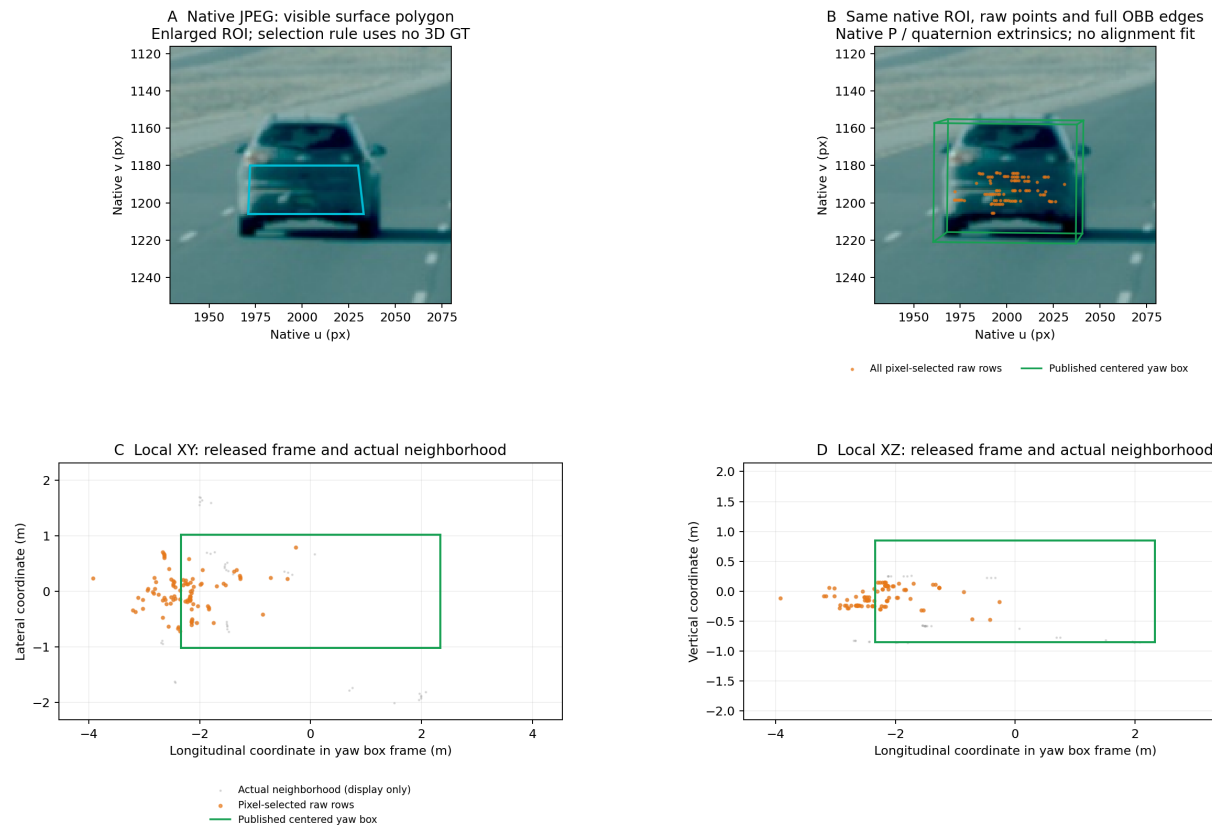
Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

F04 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

scene_36_39 | sync0032 | Vehicle:1 | Tracking_ID 1 | sample 100745 | range 93.90 m

Native ROI, released yaw box and existing point selection

F04_NEAR_MID | scene_36_39 / sync0032 / Vehicle:1 / Vehicle-Passenger
Range 93.90 m | pixel-selected N=94, beyond selected face=42, full OBB inside=51
signed face gap median -0.094 m, P05-P95 -1.209 to 0.687 m



Sensor statistics: id1: N=31, median -0.073 m, id2: N=63, median -0.118 m. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 1 (XY) / 1 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

94 rows: 51 inside full OBB; 42 beyond selected rear face; median -0.093545 m. Partial consistency only.

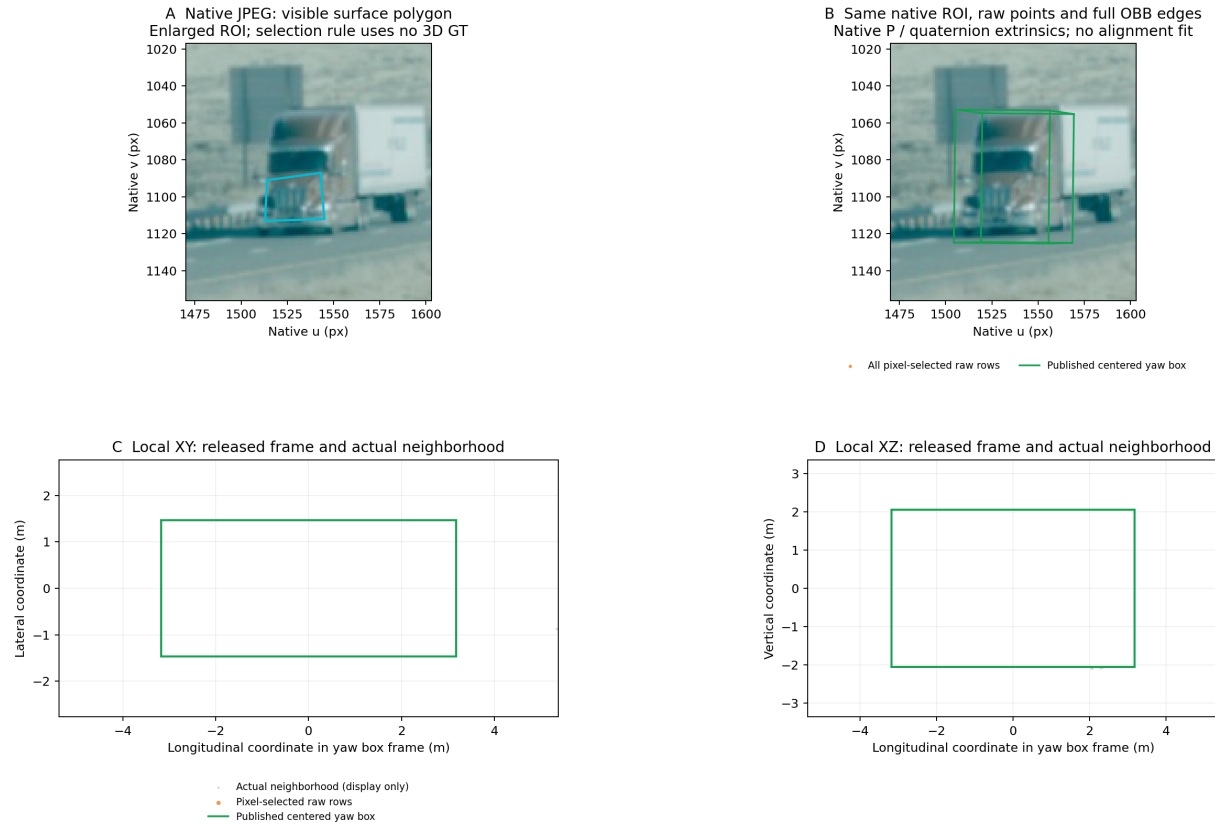
F02's selected grille has no returns; F04 is partially consistent only. No complete-agreement control is established; road/background points are not forced into a match.

F02 / Vehicle-SemiTruck-Cab / Inconclusive / projection question / partial consistency

scene_36_27 | sync0034 | Vehicle:10 | Tracking_ID 10 | sample 100601 | range 198.59 m

Native ROI, released yaw box and existing point selection

F02_FAR | scene_36_27 / sync0034 / Vehicle:10 / Vehicle-SemiTruck-Cab
Range 198.59 m | pixel-selected N=0, beyond selected face=0, full OBB inside=0
No raw return in the marked body panel; surface alignment cannot be judged



Sensor statistics: none in marked panel. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 0 (XY) / 0 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

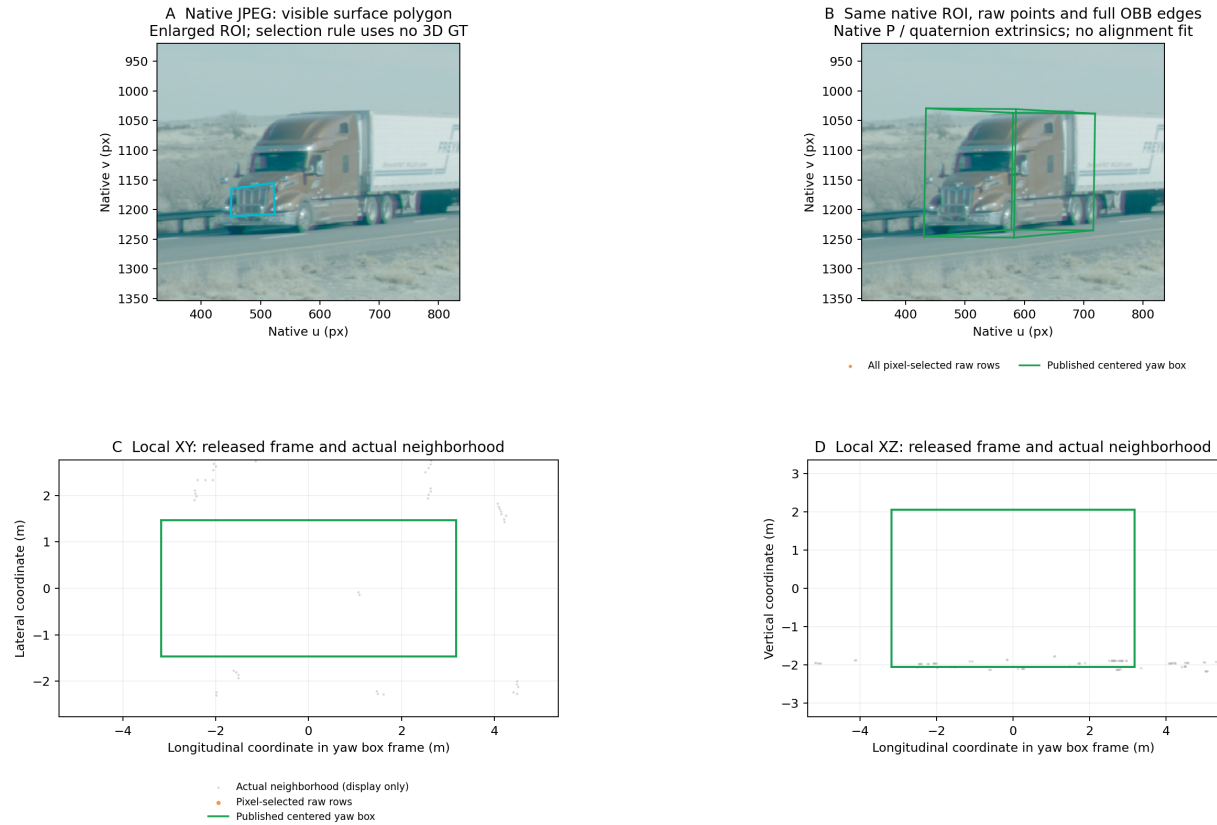
F02's selected grille has no returns; F04 is partially consistent only. No complete-agreement control is established; road/background points are not forced into a match.

F02 / Vehicle-SemiTruck-Cab / Inconclusive / projection question / partial consistency

scene_36_27 | sync0056 | Vehicle:10 | Tracking_ID 10 | sample 100623 | range 74.91 m

Native ROI, released yaw box and existing point selection

F02_NEAR_MID | scene_36_27 / sync0056 / Vehicle:10 / Vehicle-SemiTruck-Cab
Range 74.91 m | pixel-selected N=0, beyond selected face=0, full OBB inside=0
No raw return in the marked body panel; surface alignment cannot be judged



Sensor statistics: none in marked panel. Physical head mapping / time-offset reference / upstream compensation unverified.

All selected rows retained in statistics and CSV. Metric display crops 0 (XY) / 0 (XZ) selected rows outside view; no point ownership inferred from GT.

Face gap is not complete center error. Gray points are unassigned context. Each moving-object frame checked separately; no time fit or GT correction.

Existing full GT OBB point count: 2. Sparse/mixed/unassigned returns do not establish GT parameter error.

F02's selected grille has no returns; F04 is partially consistent only. No complete-agreement control is established; road/background points are not forced into a match.

H02 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

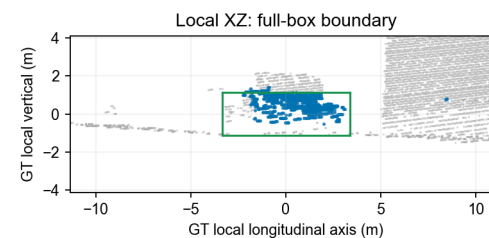
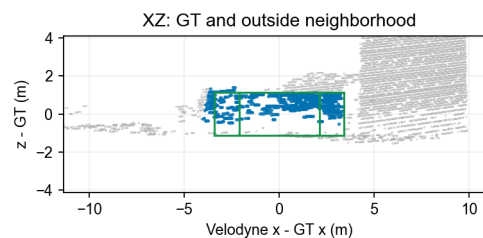
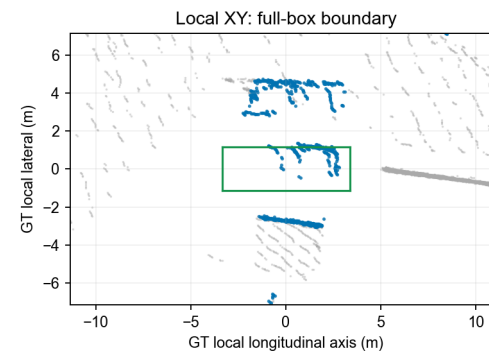
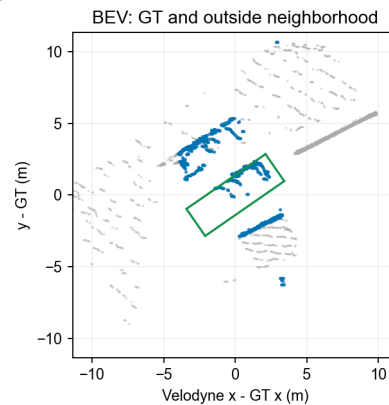
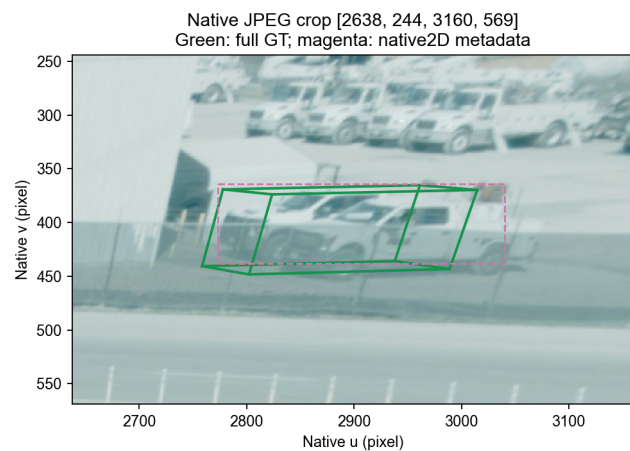
scene_42_8 | sync0178 | Vehicle:87 | Tracking_ID 87 | sample 100981 | range 61.10 m

Native camera, full GT and unassigned historical search returns

H02 | scene_42_8 / sync0178 / Vehicle:87 | Vehicle-Passenger

Range 61.10 m | GT OBB points 144 | Native-ray local points 792; outside GT 648

HISTORICAL_NEAR | HISTORICAL_C04_VEHICLE_GEOMETRY | sideward_right_front_wide



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 144. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N01 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

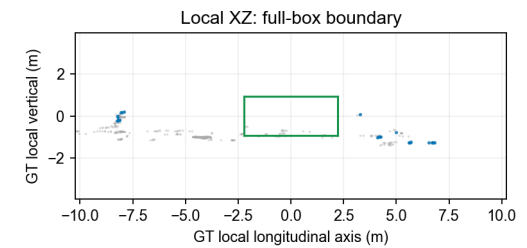
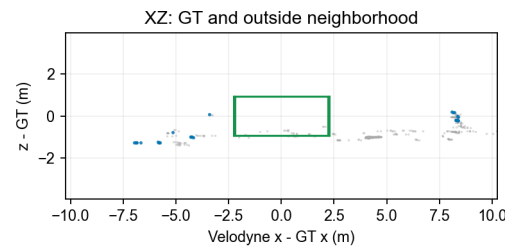
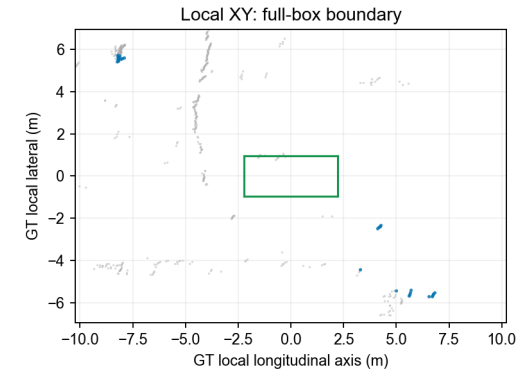
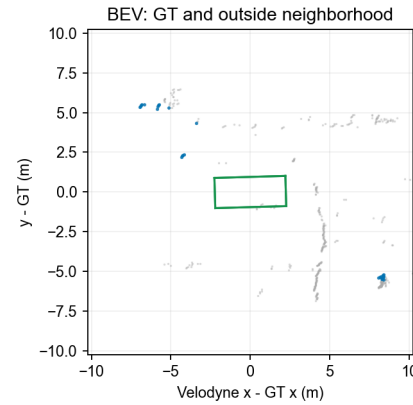
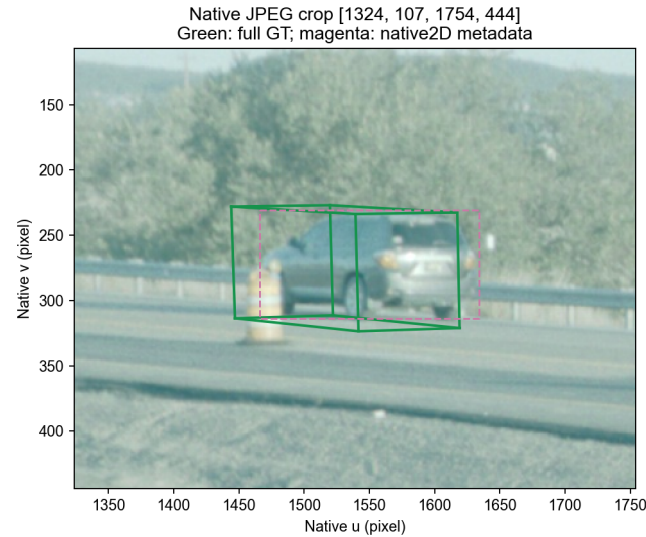
scene_28_9 | sync0285 | Vehicle:6 | Tracking_ID 6 | sample 100312 | range 73.94 m

Native camera, full GT and unassigned historical search returns

N01 | scene_28_9 / sync0285 / Vehicle:6 | Vehicle-Passenger

Range 73.94 m | GT OBB points 0 | Native-ray local points 27; outside GT 27

NEAR | FIXED_CACHED_DEV_METADATA | rearward_left_bottom_medium



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N01 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

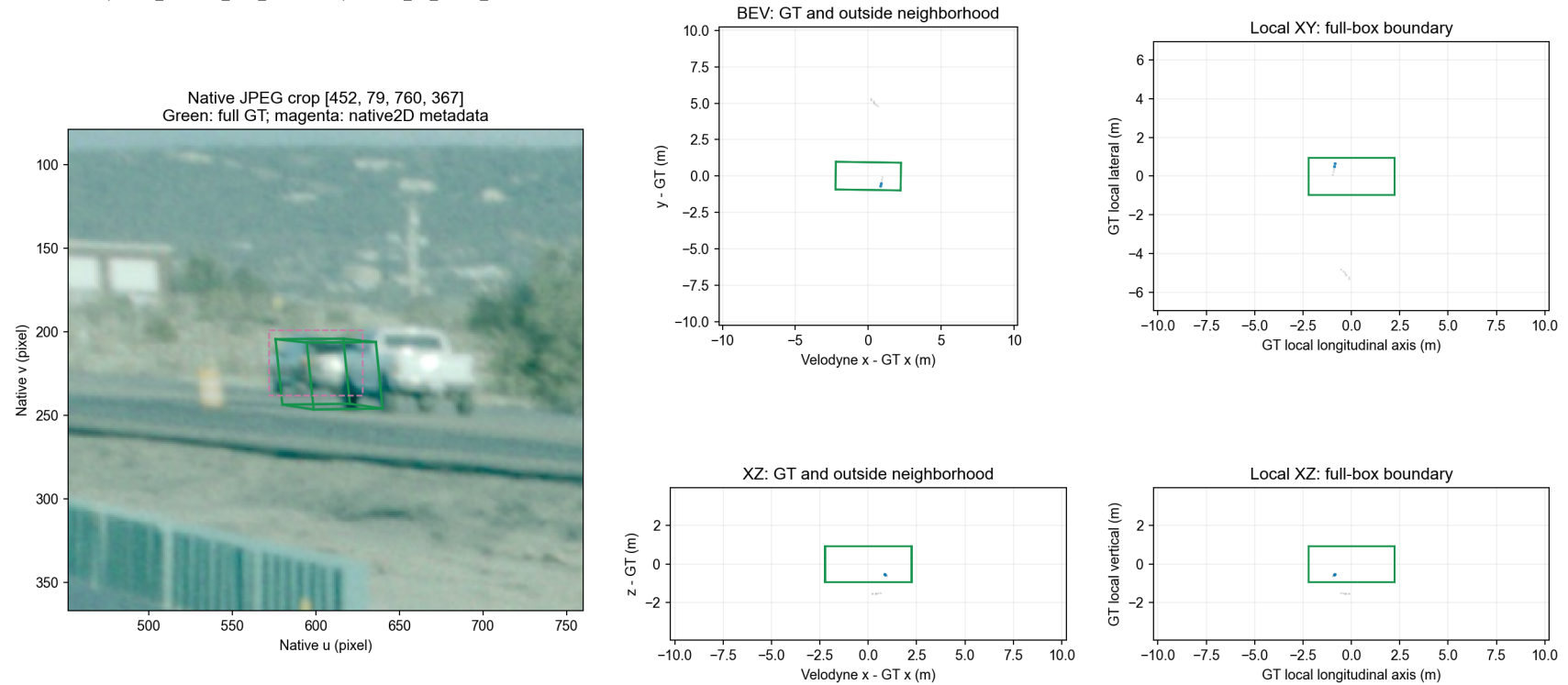
scene_28_9 | sync0324 | Vehicle:6 | Tracking_ID 6 | sample 100331 | range 176.09 m

Native camera, full GT and unassigned historical search returns

N01 | scene_28_9 / sync0324 / Vehicle:6 | Vehicle-Passenger

Range 176.09 m | GT OBB points 5 | Native-ray local points 2; outside GT 0

FAR | FIXED_CACHED_DEV_METADATA | rearward_left_bottom_medium



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 5. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N02 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

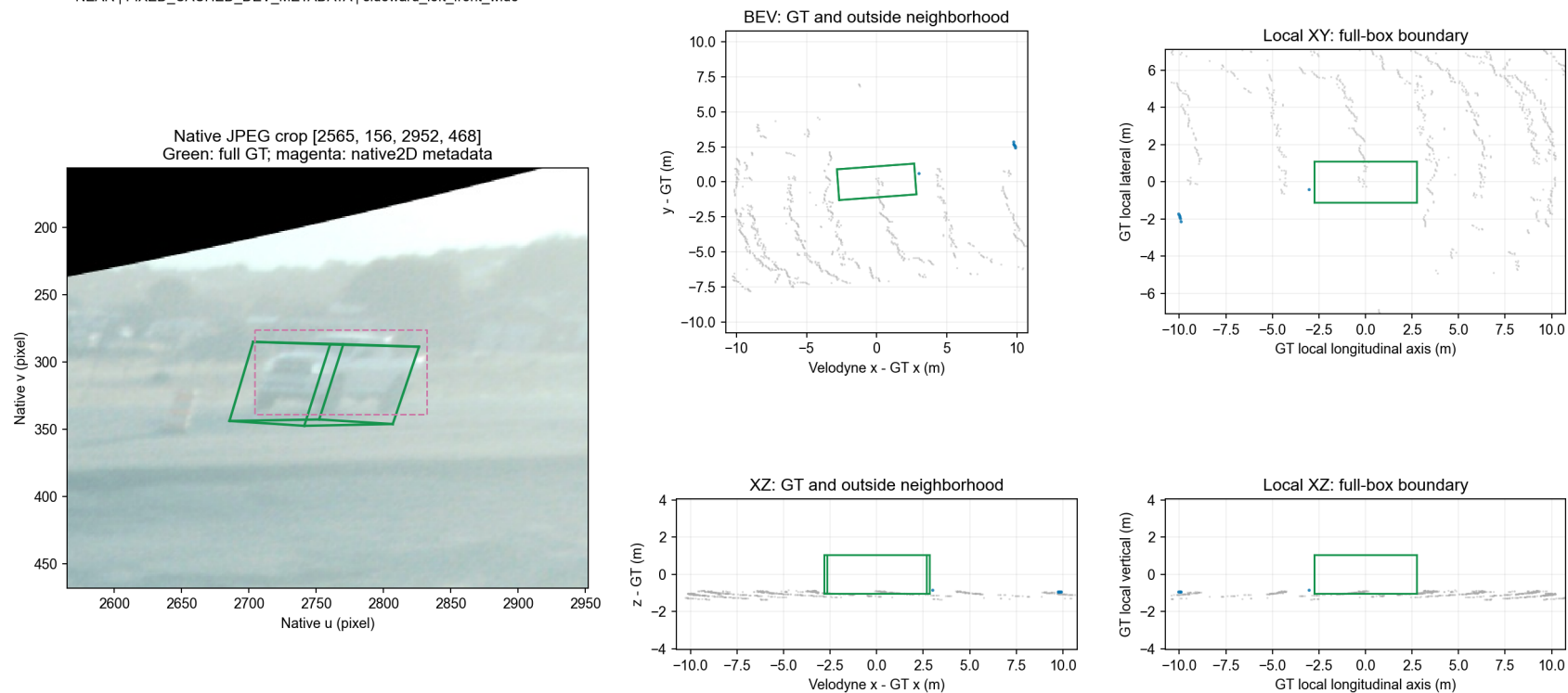
scene_28_9 | sync0253 | Vehicle:8 | Tracking_ID 8 | sample 100296 | range 74.96 m

Native camera, full GT and unassigned historical search returns

N02 | scene_28_9 / sync0253 / Vehicle:8 | Vehicle-Passenger

Range 74.96 m | GT OBB points 17 | Native-ray local points 7; outside GT 7

NEAR | FIXED_CACHED_DEV_METADATA | sideward_left_front_wide



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 17. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N02 / Vehicle-Passenger / Inconclusive / projection question / partial consistency

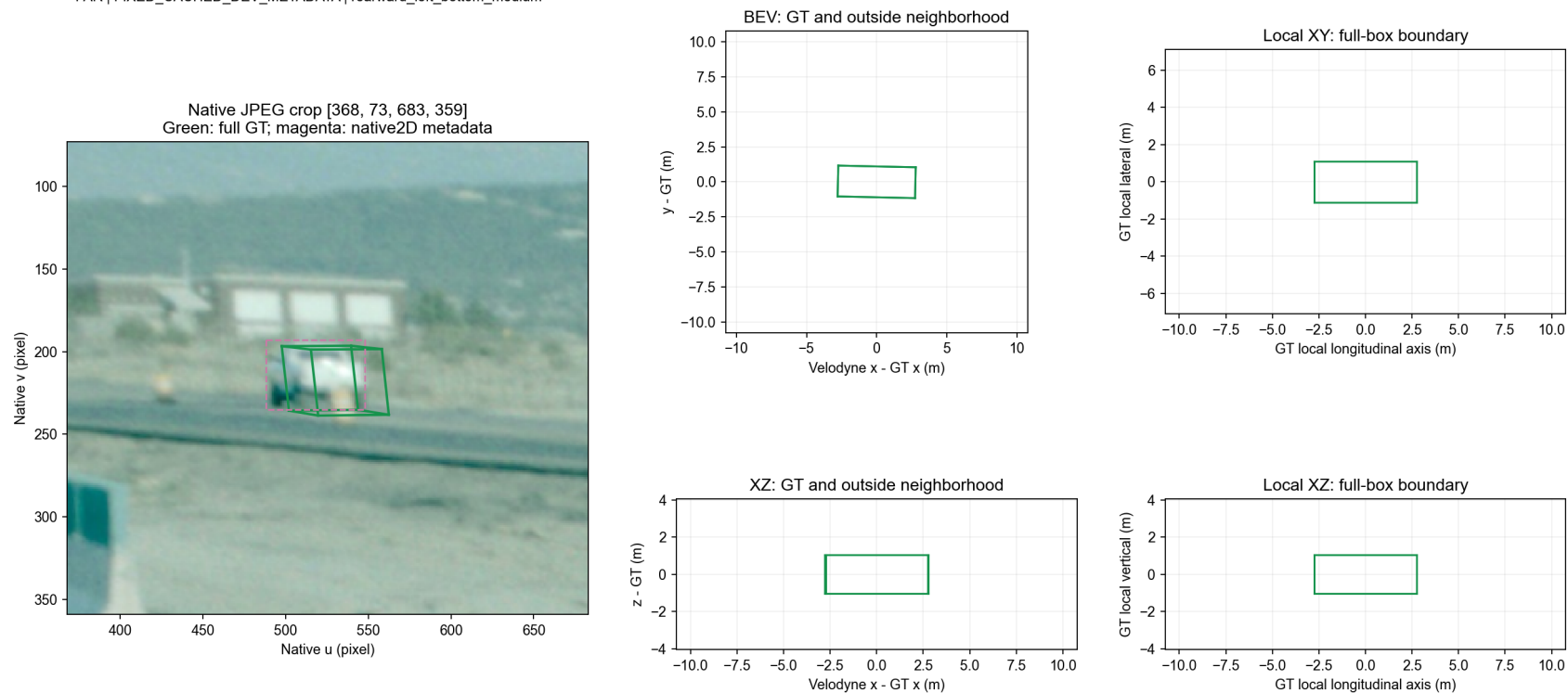
scene_28_9 | sync0340 | Vehicle:8 | Tracking_ID 8 | sample 100339 | range 201.34 m

Native camera, full GT and unassigned historical search returns

N02 | scene_28_9 / sync0340 / Vehicle:8 | Vehicle-Passenger

Range 201.34 m | GT OBB points 0 | Native-ray local points 0; outside GT 0

FAR | FIXED_CACHED_DEV_METADATA | rearward_left_bottom_medium



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N03 / Vehicle-SemiTruck-Cab / Inconclusive / projection question / partial consistency

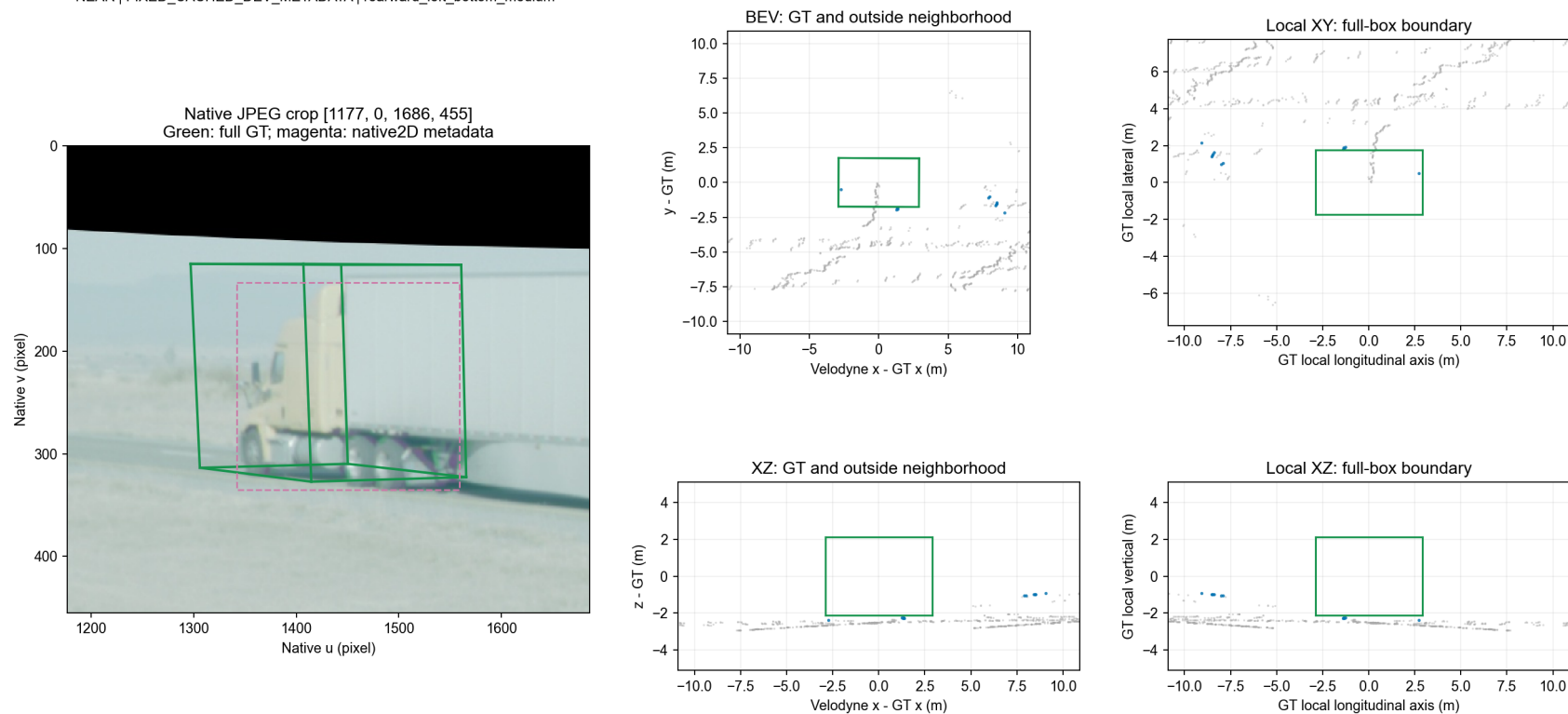
scene_28_12 | sync0086 | Vehicle:9 | Tracking_ID 9 | sample 100012 | range 72.91 m

Native camera, full GT and unassigned historical search returns

N03 | scene_28_12 / sync0086 / Vehicle:9 | Vehicle-SemiTruck-Cab

Range 72.91 m | GT OBB points 0 | Native-ray local points 12; outside GT 12

NEAR | FIXED_CACHED_DEV_METADATA | rearward_left_bottom_medium



Green: complete official GT. Gray: all actual local returns. Blue: native2D-ray local returns; target identity requires image/shape review.

Each frame inspected in its own native coordinates; no cross-frame motion subtraction, GT translation or latency compensation.

Native2D annotation origin and upstream time/deskew contracts remain unknown. Surface or boundary contradictions can still be recorded.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

N03 / Vehicle-SemiTruck-Cab / Inconclusive / projection question / partial consistency

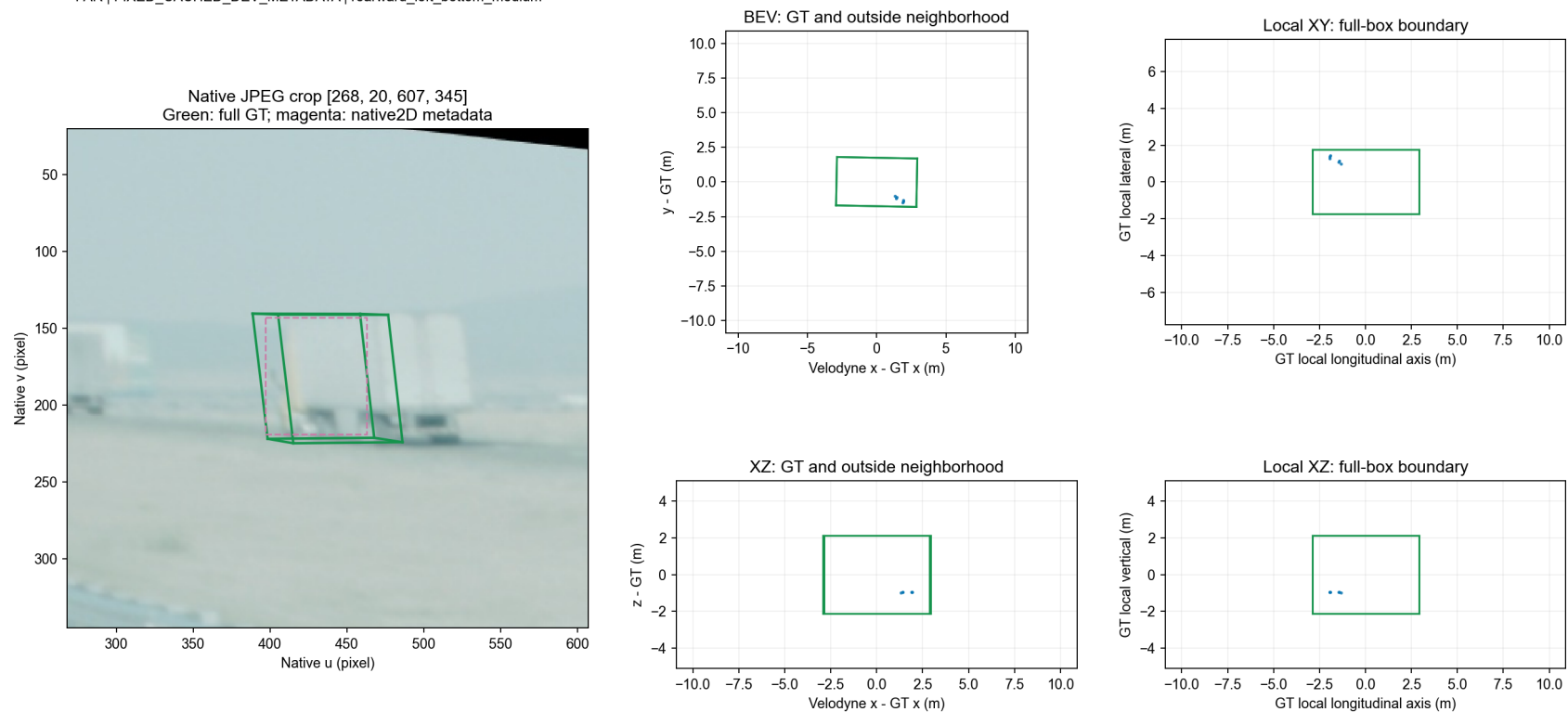
scene_28_12 | sync0126 | Vehicle:9 | Tracking_ID 9 | sample 100032 | range 196.87 m

Native camera, full GT and unassigned historical search returns

N03 | scene_28_12 / sync0126 / Vehicle:9 | Vehicle-SemiTruck-Cab

Range 196.87 m | GT OBB points 6 | Native-ray local points 6; outside GT 0

FAR | FIXED_CACHED_DEV_METADATA | rearward_left_bottom_medium



Existing full GT OBB point count: 6. Sparse/mixed/unassigned returns do not establish GT parameter error.

Gray/search returns are not automatically target points. Projected rectangle-center differences are not independently measured physical center errors.

C01 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_36_39 | sync0090 | RoadObstruction:5 | Tracking_ID 5 | sample 100803 | range 116.17 m

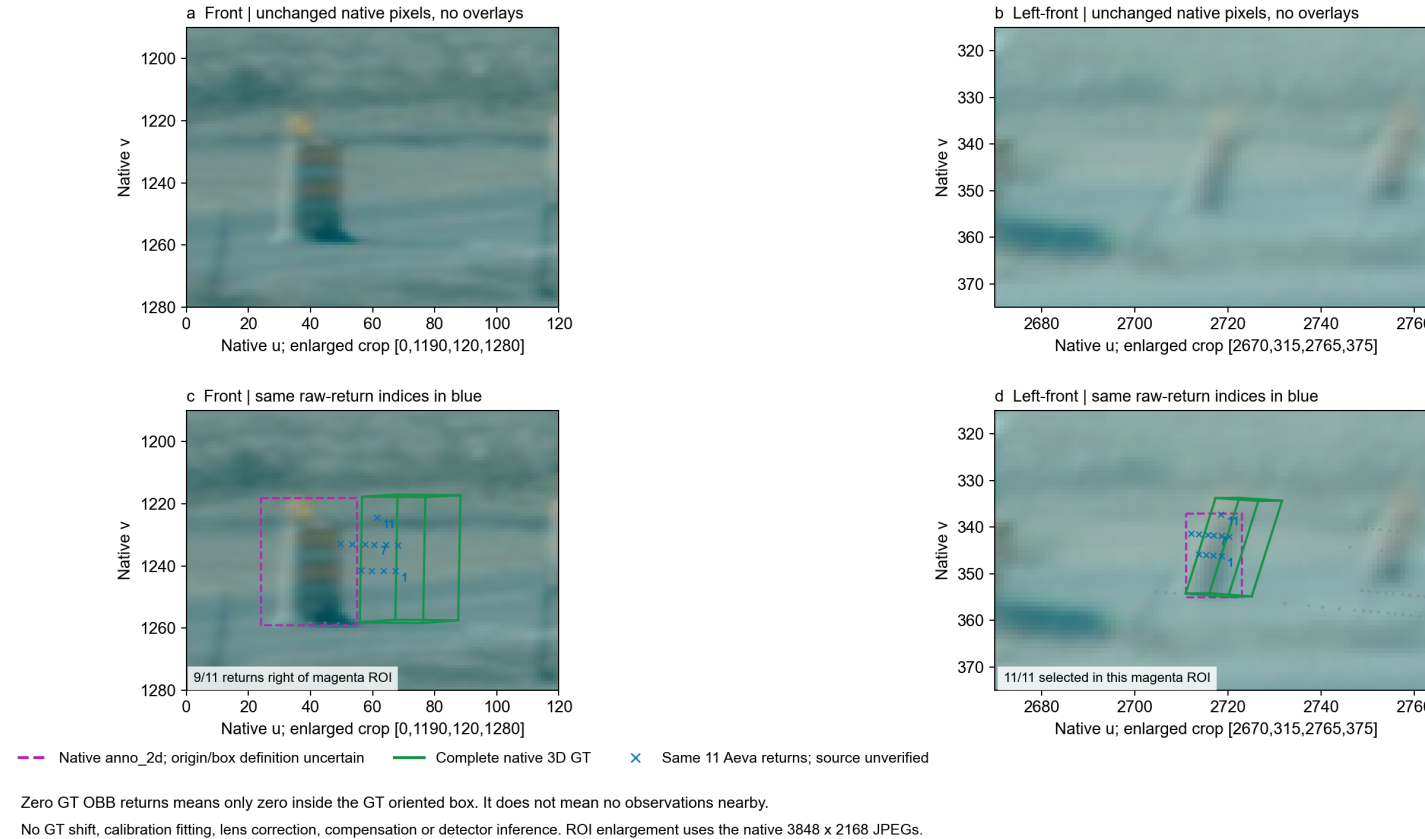
Same 11 rows projected in forward and side cameras

C01 | actual Aeva returns also show a front-view discrepancy

scene_36_39 | sync0090 | frame100803 | RoadObstruction:5 / Tracking_ID5 | Barrel | 116.17 m

Same 11 raw returns: selected in the left-view native 2D ROI; 9 project to the RIGHT of the front-view 2D ROI.

This supports a front-view projection/correspondence issue; the physical point source and the exact cause remain unverified.



Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

The side-image rectangle defines selection; 9 points project to the right of the visible forward-image barrel box. This supports a correspondence question; physical attribution is unverified.

C01 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_36_39 | sync0090 | RoadObstruction:5 | Tracking_ID 5 | sample 100803 | range 116.17 m

Existing preprojection geometry of the same 11 points

C01 | the GT and fixed returns are already separated BEFORE camera projection

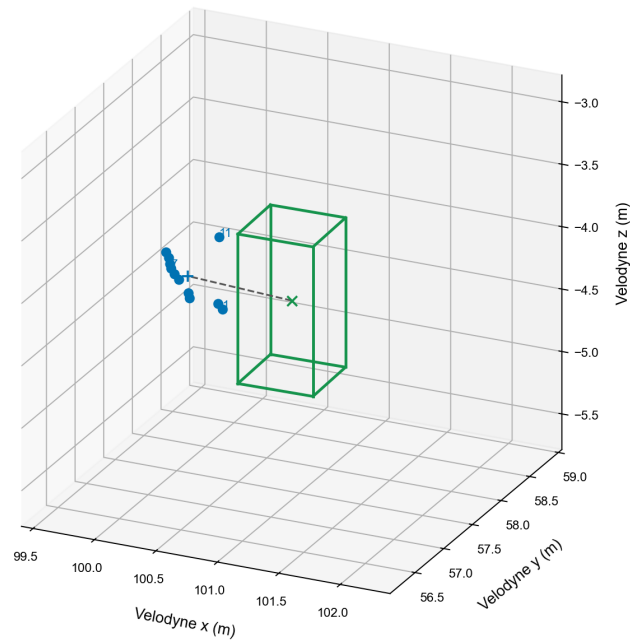
scene_36_39 | sync0090 | frame100803 | RoadObstruction:5 / Tracking_ID5 | all coordinates in Velodyne, metres

GT center (100.905686, 57.563654, -4.284547); dimensions 0.604884 x 0.601146 x 1.197239 m; yaw 0.02455810 rad

11-point mean - GT center = (-0.794828, -0.109655, +0.097014) m; mean-to-center distance 0.808201 m

Point mean is only a cluster descriptor. Suspected Barrel-surface returns are NOT independently certified.

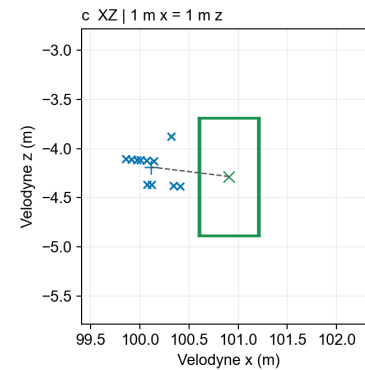
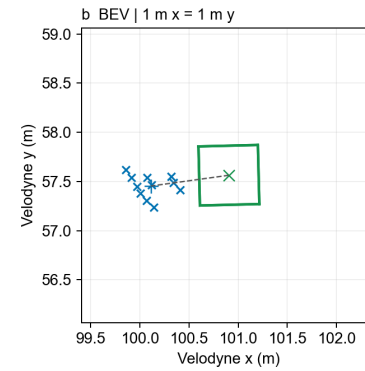
a Complete yaw-oriented GT | orthographic 3D; equal metric axis scales



— Native complete GT; green x = geometric center x Same fixed 11 raw returns; source unverified + Cluster mean descriptor (not object center)

11/11 outside GT OBB; point-to-GT solid gap 0.196365-0.743826 m; point-to-GT center distance 0.526308-1.064496 m.

Pure camera projection error cannot create this pre-projection 3D separation. Point identity / full object extent are still unverified.



Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

All points outside the GT OBB are a local clue. Cluster statistics do not substitute for the complete barrel center.

C01 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_36_39 | sync0090 | RoadObstruction:5 | Tracking_ID 5 | sample 100803 | range 116.17 m

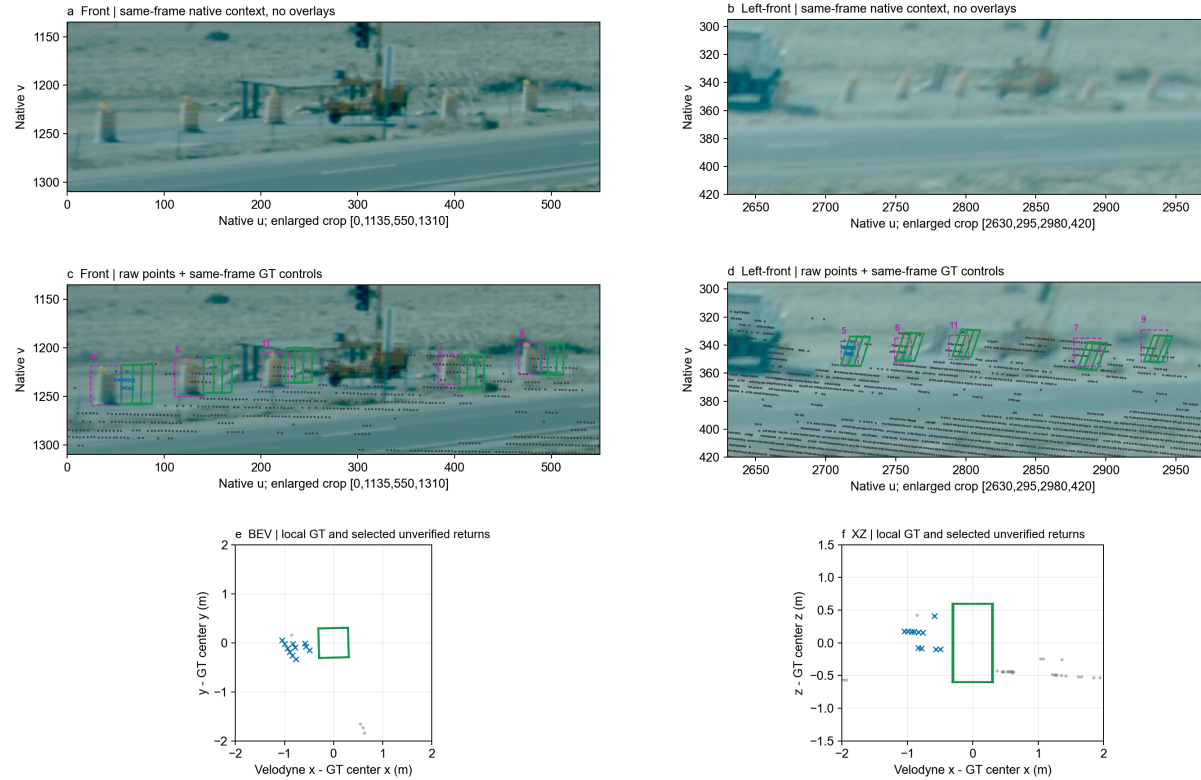
Existing same-frame projection controls

C01 | same-frame controls and local geometry; no new sample experiment

Five visible Barrel records: projected 3D AABB center minus native2D center, horizontal offset (front / left, px)

IDs 5: +32.6 / +4.3 6: +30.1 / +4.8 11: +25.3 / +5.4 7: +23.6 / +3.8 9: +22.7 / +2.8

Front-view offsets are shared by several nearby objects. Raw point clusters also sit to the right of visible barrels.



Native anno_2d Native GT Actual returns, all depths; no occlusion test Left ray-ROI returns

All 11 selected returns lie outside GT; solid gap 0.196-0.744 m. Surface identity/full extent is not independently certified.

Gray: all 94 returns in XY6m neighborhood (cropped metric viewport below). Blue: same 11 raw indices; no inferred object center.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Other same-frame barrels are auxiliary projection controls; they add no cases and do not certify GT parameter error.

C02 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_28_9 | sync0346 | RoadObstruction:22 | Tracking_ID 22 | sample 100342 | range 73.51 m

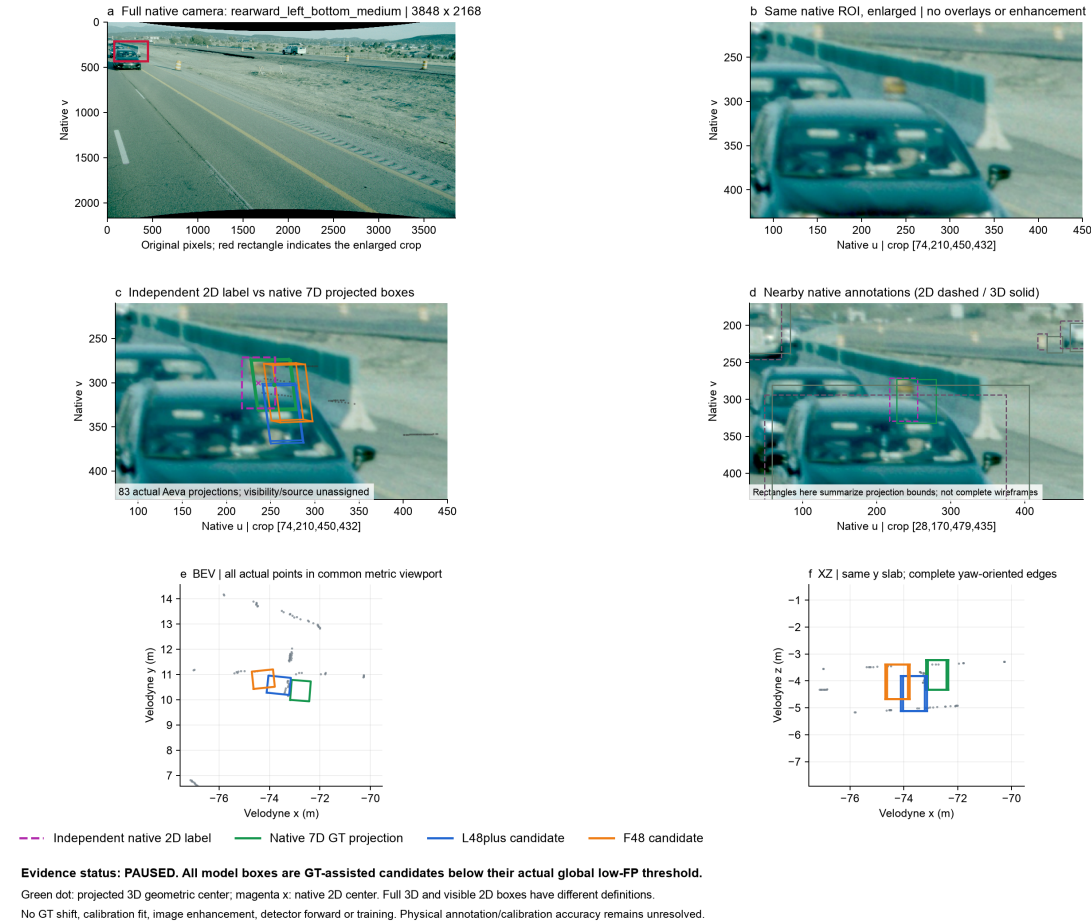
Native projection review; historical model boxes are auxiliary, not truth

C02 review v1 | native annotation / projection discrepancy

scene_28_9 | sync 0346 | frame 100342 | RoadObstruction:22 / RoadObstruction-Barrel | 73.51 m

Projected 8-corner AABB center - native 2D center: (+17.62, +2.49) px | GT points 0 | UNRESOLVED

Native GT 7D exact to frozen data; source row 10, frozen GT index 10. Aeva - camera = 27.975 ms.



Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Zero OBB points do not imply no neighborhood returns. GT-assisted candidate identity is unconfirmed and does not prove inference of a particular barrel.

C05 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_28_9 | sync0107 | RoadObstruction:18 | Tracking_ID 18 | sample 100223 | range 187.66 m

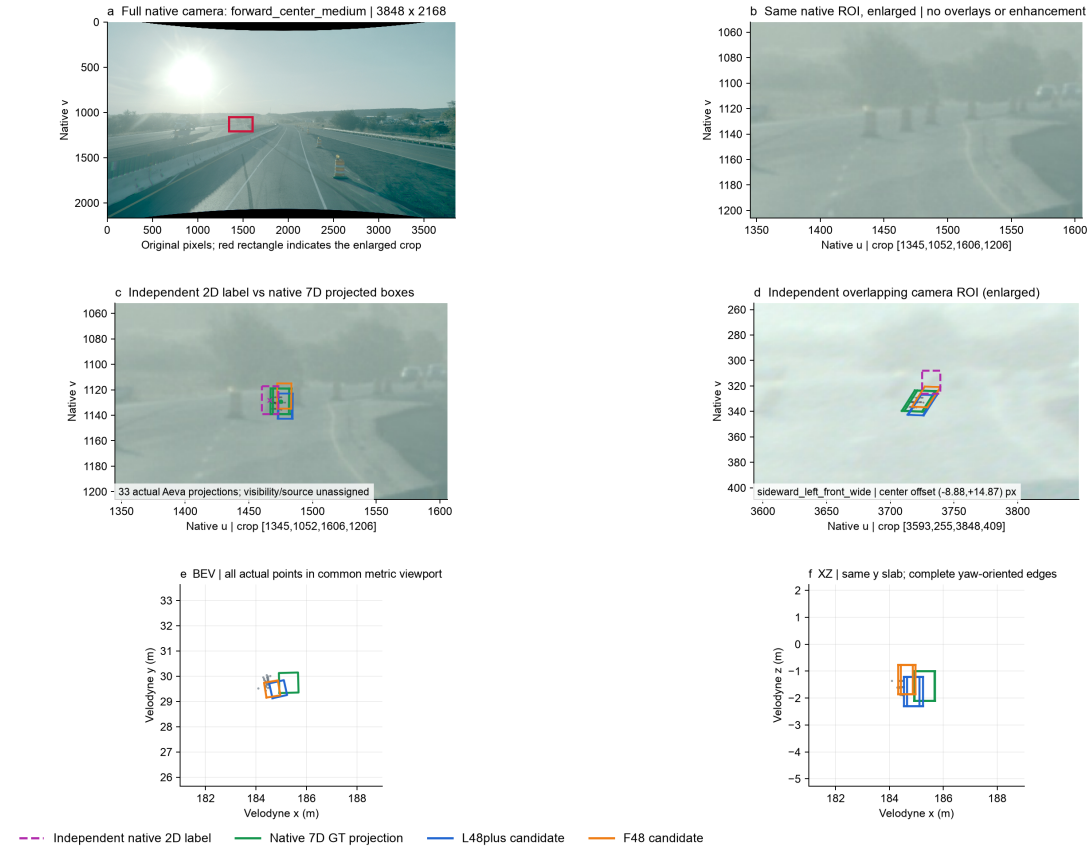
Native projection review; historical model boxes are auxiliary, not truth

C05 review v1 | native annotation / projection discrepancy

scene_28_9 | sync 0107 | frame 100223 | RoadObstruction:18 / RoadObstruction-Barrel | 187.66 m

Projected 8-corner AABB center - native 2D center: (+8.73, +1.10) px | GT points 0 | UNRESOLVED

Native GT 7D exact to frozen data; source row 9, frozen GT index 9. Aeva - camera = 27.898 ms.



Evidence status: PAUSED. All model boxes are GT-assisted candidates below their actual global low-FP threshold.

Green dot: projected 3D geometric center; magenta x: native 2D center. Full 3D and visible 2D boxes have different definitions.

No GT shift, calibration fit, image enhancement, detector forward or training. Physical annotation/calibration accuracy remains unresolved.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Zero OBB points do not imply no neighborhood returns. GT-assisted candidate identity is unconfirmed and does not prove inference of a particular barrel.

C05 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_28_9 | sync0107 | RoadObstruction:18 | Tracking_ID 18 | sample 100223 | range 187.66 m

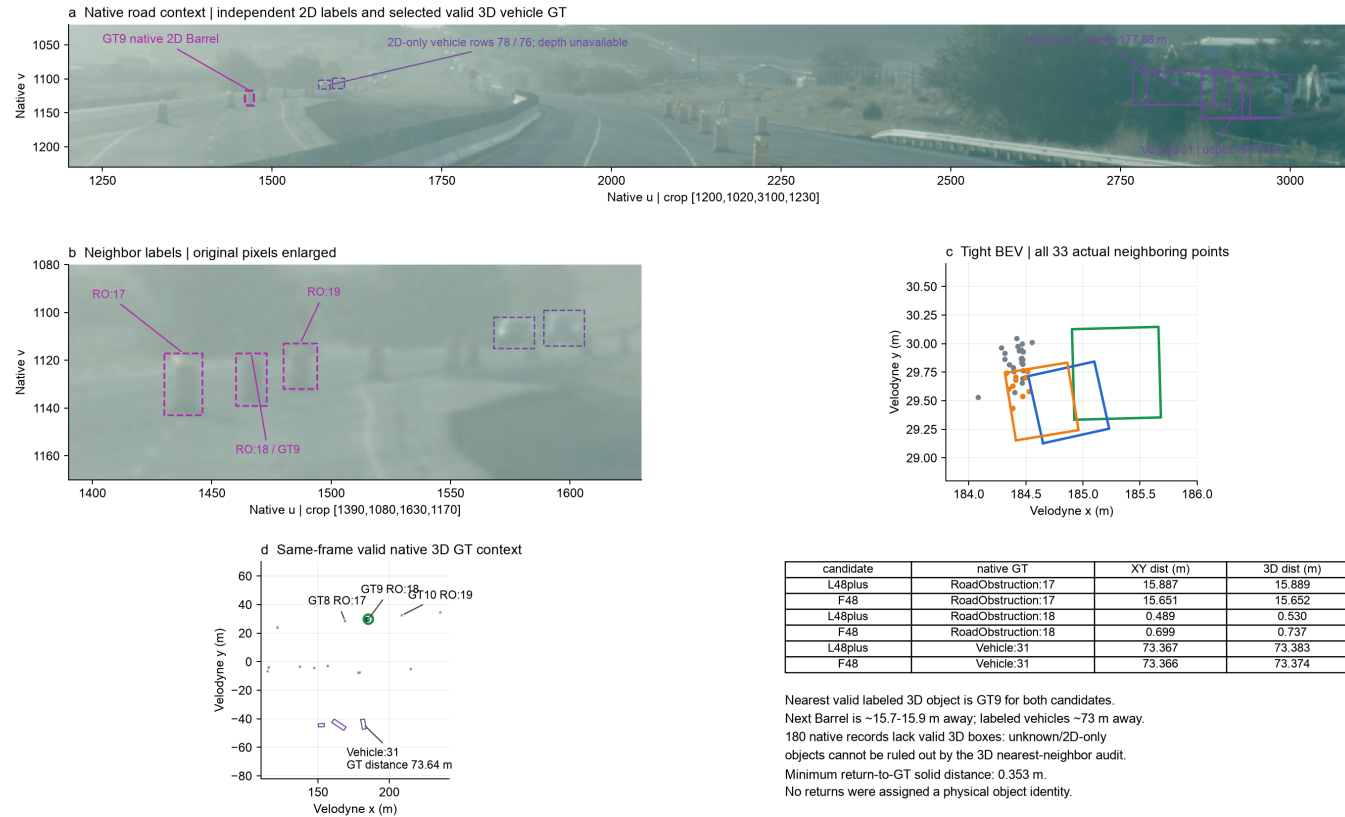
Unassigned neighborhood coordinates and historical candidates

C05 neighbor audit v1 | numerical proximity does not prove identity

scene_28_9 | sync 0107 | frame 100223 | GT9 RoadObstruction:18 / Barrel | 187.66 m

33 nearby actual points: GT contains 0; L48plus q20 contains 0; F48 q15 contains 10.

Both candidates decode as RoadObstruction (~0.6 x 0.6 x 1.08 m); physical point source remains UNRESOLVED.



Withdrawn inference: "the model inferred this particular zero-point Barrel". Retained: GT-assisted candidate geometry and exact point counts.

Native 2D dashed magenta; 3D GT green; L48plus blue; F48 orange; vehicles purple. Depth = camera center z for valid 3D GT only.

Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Displayed point IDs are local indices; pointwise raw indices were not saved. Candidates and neighborhood clusters are not barrel truth.

C06 / RoadObstruction-Barrel / Inconclusive / projection question / partial consistency

scene_36_39 | sync0057 | RoadObstruction:7 | Tracking_ID 7 | sample 100770 | range 219.66 m

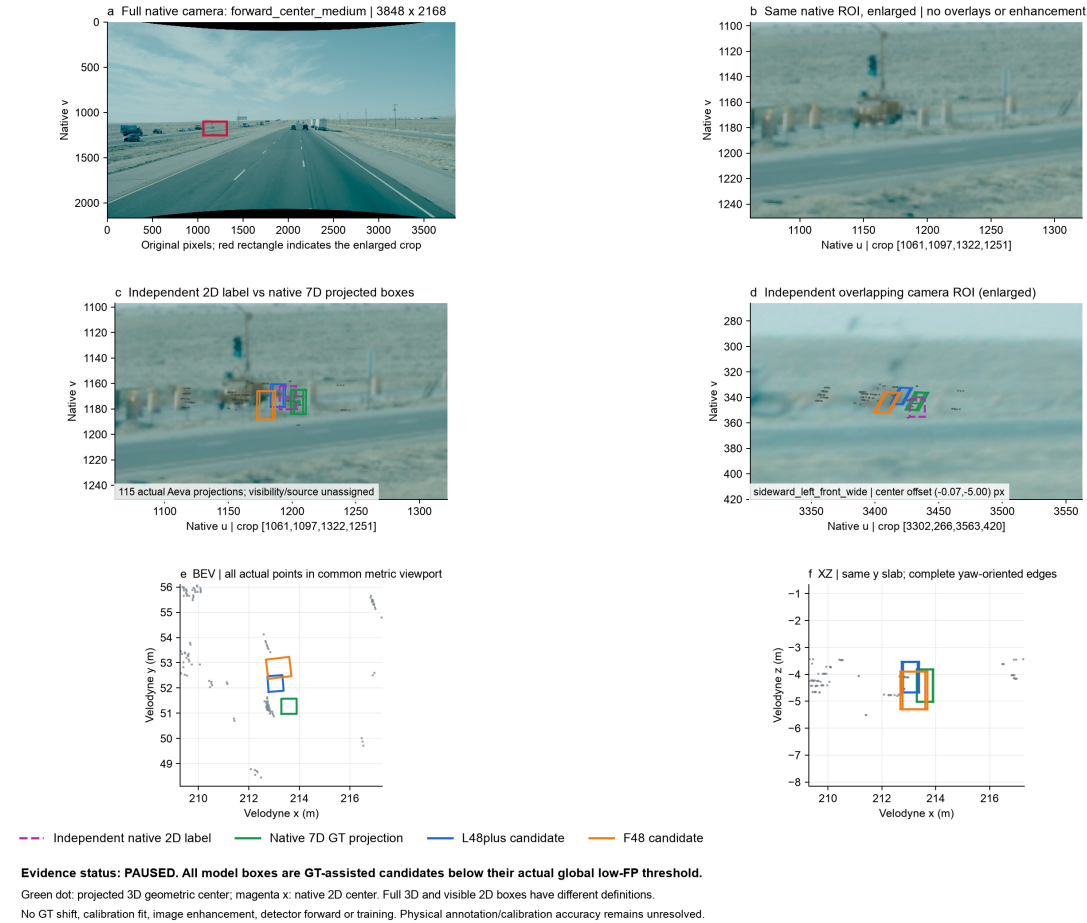
Native projection review; historical model boxes are auxiliary, not truth

C06 review v1 | native annotation / projection discrepancy

scene_36_39 | sync 0057 | frame 100770 | RoadObstruction:7 / RoadObstruction-Barrel | 219.66 m

Projected 8-corner AABB center - native 2D center: (+8.60, +3.54) px | GT points 0 | UNRESOLVED

Native GT 7D exact to frozen data; source row 17, frozen GT index 17. Aeva - camera = 27.929 ms.



Existing full GT OBB point count: 0. Sparse/mixed/unassigned returns do not establish GT parameter error.

Zero OBB points do not imply no neighborhood returns. GT-assisted candidate identity is unconfirmed and does not prove inference of a particular barrel.